

Software Engineering Document for TicketIQ

Client: T-Mobile

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Introduction of Document – Purpose of this Document, Audience, and Structure – 02/10/2026

The purpose of this Software Engineering Document (SED) is to define the design, requirements, architecture, and testing strategy for the TicketIQ System. TicketIQ is an AI-powered IT Helpdesk ticket management and prioritization system. This basic functionality of this system is that it allows T-Mobile users to submit help tickets, which are then analyzed by a Large Language Model (LLM) through Natural Language Processing (NLP) to determine the severity and urgency, which are used to determine the priority, which is then placed in a priority queue for an assigned IT technician

This SED provides specifications for the systems' functional and non-functional requirements, site maps, use case diagrams, wireframes, architecture design, implementation methods, system testing plan, and software evolution specifications. This SED is fundamental to the development of TicketIQ and everything that needs to be included.

This SED is intended for stakeholders involved with the development and approval of the TicketIQ system. These stakeholders include, but are not limited to, T-Mobile Software Developers, T-Mobile Software Architects, T-Mobile Requirements Engineers, T-Mobile Project Managers, T-Mobile Testing Engineers, T-Mobile Quality Assurance Engineers, T-Mobile Board Members, T-Mobile Security Engineers, and T-Mobile Cloud Engineers.

In order to give a logical flow between the concept requirements and the technical implementation, the document is organized into a series of major sections, which include: Requirements and System Modeling, which contains detailed stakeholder interviews and functional requirements; User Stories which define the system value as viewed by various perspectives; System Design which contains site maps and other technical drawings and finally, Implementation, Software Testing and Evolution sections which guide the system throughout its entire lifecycle.

Requirements / System Modeling – 02/10/2026

Gathering requirements

The approach to gathering requirements for the TicketIQ system was through a structured requirement engineering process. This included stakeholder identification, stakeholder interviews, a fundamental understanding and analysis of the Statement of Work (SOW), and the creation of user stories. The TicketIQ team, consistent with Dhruv Yadav, Nicolas Gaffney, Azzam Zahid, and Rinin Abraham, first compiled a list of 40+ stakeholders who are directly involved in the development and approval of the TicketIQ system. Next, the team conducted stakeholder interviews to better understand the current workflow, common challenges, and understandings needed to ensure that TicketIQ is conceptually and architecturally sound and everything is approved for development.

Additionally, the team discussed how users should interact with the system and which features they would like to have. Furthermore, through these interviews and user personas, the team was able to translate these responses into 40+ Functional Requirements, 16+ Non-Functional Requirements, 16 user stories, 4 Use Case Diagrams, 4 Sequence Diagrams, and 12 Wireframe Designs that address all aspects of the stakeholder's answers.

Stakeholder Table & Interview Questions

Stakeholder	Company	Questions	Answer	Student/Owner
System Administrator	T-Mobile IT Operations	What monitoring tools are needed to track system health?	“We need monitoring for server uptime, API failures, database status, and ticket processing latency. Alerts should trigger if any critical service goes down.”	Dhruv Yadav
Cloud Infrastructure Engineer	T-Mobile Cloud Engineering	What cloud architecture should support high ticket volumes?	“The system should use scalable cloud infrastructure with separate services for ticket processing,	Dhruv Yadav

			classification, and analytics.”	
Security Engineer	T-Mobile Cybersecurity	What authentication and security policies should be implemented?	“Users should authenticate through T-Mobile SSO. Access should be role-based and system activity should be logged.”	Dhruv Yadav
DevOps Engineer	T-Mobile Platform Engineering	What CI/CD tools should be used for automated deployment?	“The system should support automated build, testing, and deployment pipelines with rollback capability.”	Dhruv Yadav
Network Engineer	T-Mobile Network Engineering	What network incidents should automatically be classified as high severity?	“Network outages affecting multiple users or locations should automatically be classified as high severity.”	Dhruv Yadav
Infrastructure Engineer	T-Mobile Infrastructure Team	What infrastructure failures should trigger urgent tickets?	“Server outages, authentication failures, and database outages should be treated as urgent incidents.”	Dhruv Yadav
IT Operations Engineer	T-Mobile IT Operations	What ticket information helps diagnose issues quickly?	“Tickets should include system name, error messages, affected location, and issue description.”	Dhruv Yadav

Data Engineer	T-Mobile Data Engineer	What ticket data should be stored for analytics?	“We should store severity, urgency, timestamps, technician assignment, and resolution time.”	Dhruv Yadav
Machine Learning Engineer	T-Mobile AI/ML Team	What training data improves ticket classification accuracy	“Historical ticket data and corrected classifications should be used to improve model accuracy.”	Dhruv Yadav
Enterprise Architect	T-Mobile Enterprise Architecture	What system must TicketIQ integrate with?	“The system should integrate with the existing ticket system, SSO authentication, and employee profile data.”	Dhruv Yadav
T-Mobile Security Team	T-Mobile Information Security	How sensitive is the data stored, and how should it be classified?	Tickets may contain PII and internal system details, so they must be classified as confidential. All data at rest must be encrypted using AES-256, and access must be logged and auditable.	Nicolas Gaffney
		How will you test the security of TicketIQ before deploying to production?	We require a third-party penetration test covering OWASP Top 10	Nicolas Gaffney

			vulnerabilities, a review of all API authentication mechanisms, and a full RBAC audit to confirm no privilege escalation paths exist.	
		How long should SSO tokens be valid?	SSO tokens should expire after 8 hours of inactivity. All sessions must be invalidated server-side on logout, and any token reuse after expiry must be blocked at the API gateway level.	Nicolas Gaffney
		Should a data breach occur, what steps does TicketIQ need to take	The system must be able to trigger an automated alert to our SIEM platform within 5 minutes of detecting anomalous access patterns. A full audit trail must be preserved and handed off to our IR team immediately.	Nicolas Gaffney
T-Mobile Data Analytics Team	T-Mobile Business Intelligence	How should the reports be formatted?	We primarily consume data via CSV exports and REST API feeds into our	Nicolas Gaffney

			Tableau dashboards. JSON over HTTPS is preferred for real-time feeds; CSV works fine for scheduled weekly exports.	
		How often does the data need to refresh for the administration?	Near-real-time for executive dashboards — ideally a 15-minute refresh cycle. For weekly trend reports, a nightly batch job is acceptable. Historical data going back at least 12 months should be queryable.	Nicolas Gaffney
		What besides volume and resolution time does TicketIQ need to track?	Yes — we need first-contact resolution rate, SLA breach rate broken down by IT worker and ticket category, and AI classification accuracy over time. Segmentation by region and department is also important.	Nicolas Gaffney
AWS Cloud Operations Team	T-Mobile Cloud Infrastructure	What are the estimated storage and bandwidth requirements?	We estimate roughly 500 GB of ticket data in year one, growing ~20% annually. There	Nicolas Gaffney

			are existing S3 buckets with lifecycle policies we can repurpose.	
		What is the approved AWS region for deployment, and are there redundancy requirements we must meet?	All production workloads must deploy in us-east-1 with active failover to us-west-2. Multi-AZ is mandatory for any RDS or DynamoDB table. ECS services must have a minimum of 2 tasks running across availability zones at all times.	Nicolas Gaffney
		How is the cost controlled, and what should Bettertech do in the case of unexpected surges in spending?	We use AWS Budgets with a hard alert at 110% of the monthly baseline. If spend is projected to exceed that, BetterTech must notify our FinOps team within 24 hours. Auto-scaling policies must have a maximum instance cap configured to prevent runaway costs.	Nicolas Gaffney

Frontend Developer	UX Design	What color scheme and branding should be used?	The interface should follow the company's branding guide using its signature magenta and white color scheme.	Rinin Abraham
QA Engineer	Software Quality	What browser support is acceptable for the portal used for the helpdesk?	The portal must be up to date and fully functional on modern browsers like Chrome, Firefox, Safari, and Edge.	Rinin Abraham
API Integration Specialist	Platform Team	What protocol should be used for the classification feed?	Must be delivered via JSON over HTTPS for quick dashboard updates	Rinin Abraham
Compliance Officer	Legal & Privacy	How must user PII be handled using ticket logs?	All PII must be masked within logs and must abide by the privacy and data handling policies	Rinin Abraham
Support Team Lead	IT Support	How should the algorithm prioritize tickets?	Should prioritize tickets based on a technician's previous resolution categories and their areas of expertise.	Rinin Abraham
System Architect	Platform Engineering	What is the requirement for mobile accessibility?	Should provide a web interface that is accessible on both desktop and mobile.	Rinin Abraham

Database Administrator	Database Services	What is the retention policy for ticket reclassification?	A complete log of status changes must be kept for at least 12 months for audit.	Rinin Abraham
DevOps Engineer	Platform Engineer	What is the recovery time objective for the failover?	The active failover to us-west-2 should be triggered within 15 minutes of a primary region outage detection.	Rinin Abraham
User Experience Researcher	Internal Tools	What status do end users need to see?	They need to view their ticket status and classification so they can track progress.	Rinin Abraham
Backup Administrator	Infrastructure	How often should metadata be backed?	They should be performed nightly and stored across different AWS regions.	Rinin Abraham
IT Support Employee	T-Mobile IT Support	What information do you need in a ticket to quickly triage severity and route it correctly?	Ticket should include. Affected system/app, location/region, error message or screenshots, impact scope, and what troubleshooting has been attempted.	Azzam Zahid

IT Support Specialist	T-Mobile IT Support	What actions should be available for escalated tickets (claim/assign/escalate further)?	Support should be able to filter escalated tickets, claim ownership, assign to a specialist/team, and add internal notes; escalations should be visible in ticket history.	Azzam Zahid
IT Manager	T-Mobile IT Operations	When a high severity ticket is unassigned, when should the system auto-escalate, and who should be notified?	Auto escalate if unassigned beyond the threshold (ex. 30 mins). Notify the on-call lead and optionally a backup group; include ticket link, severity, age, and impacted service.	Azzam Zahid
Helpdesk Operations Manager	T-Mobile IT Operations	What KPIs should weekly reports include for escalations and SLA health?	Weekly reports should include escalation count, time to assignment, time to resolve, SLA breach rate, top categories, and breakdown by team/region.	Azzam Zahid
System Administrator	T-Mobile IT Operations	What admin controls are needed to manage escalation rules safely?	Admin should configure time thresholds, severity thresholds, and notification targets; changes should be	Azzam Zahid

			versioned and auditable with rollback capability.	
Security Engineer	T-Mobile Cybersecurity	What should be logged for escalation/override actions for security review?	Log ticketId, actor (system/admin), timestamp, old/new severity/status, and reason. Admin overrides must be auditable; suspicious access should trigger alerts to security tools.	Azzam Zahid
IAM/SSO Administrator	T-Mobile Identity & Access	What authentication/authorization requirements apply to escalation overrides and rule configuration?	Use SSO for authentication; enforce RBAC so only admins can override/configure; sessions should expire after inactivity and privileged actions should require re-auth if needed.	Azzam Zahid
DevOps/SRE Engineer	T-Mobile Platform Engineering	How often should the escalation job run and what reliability requirements matter?	Run frequently (ex: every 5 minutes). Job failures should alert SRE; the system must avoid missed runs and support retries without duplicating escalation actions.	Azzam Zahid

Data/Analytics Engineer	T-Mobile Business Intelligence	What data format and refresh rate are needed for escalation of analytics dashboards?	Provide REST/JSON for near real time dashboards and CSV for scheduled exports; keep at least 12 months of history for trend analysis and audit queries.	Azzam Zahid
Compliance/Privacy Officer	T-Mobile Legal & Privacy	How should sensitive data and PII be handled in tickets and logs?	Treat ticket content as confidential; mask PII in logs and reports, enforce retention rules, encrypt data at rest and in transit, and limit access by role.	Azzam Zahid

Functional Requirements

Number	Functional Requirements (FR)	Student/Owner
FR - 01	The system shall allow T-Mobile users to submit IT helpdesk tickets.	Dhruv Yadav
FR - 02	The system shall require tickets to include a problem description and affected system or service.	Dhruv Yadav
FR - 03	The system shall analyze ticket descriptions using Natural Language Processing (NLP) to determine ticket severity and urgency.	Dhruv Yadav
FR - 04	The system shall classify tickets into severity levels such as low, medium, and high based on the issue description.	Dhruv Yadav
FR - 05	The system shall automatically determine ticket urgency based on incident type and system impact.	Dhruv Yadav
FR - 06	The system shall automatically prioritize tickets based on their severity and urgency classification.	Dhruv Yadav
FR - 07	The system shall display a prioritized ticket queue for technicians to view and manage assigned tickets.	Dhruv Yadav
FR - 08	The system shall allow technicians to search and filter tickets using keywords, severity level, or ticket category.	Dhruv Yadav

FR - 09	The system shall authenticate users through T-Mobile's Single Sign-On (SSO) system.	Dhruv Yadav
FR - 10	The system shall store ticket metadata including severity, urgency, timestamps, and technician assignment for analytics.	Dhruv Yadav
FR - 11	The system shall provide a login interface allowing T-Mobile users to enter credentials to access TicketIQ.	Dhruv Yadav
FR - 12	The system shall display an error message when login credentials are invalid.	Dhruv Yadav
FR - 13	The system shall allow users to select an issue category when submitting a ticket.	Dhruv Yadav
FR - 14	The system shall allow users to upload attachments when submitting a ticket.	Dhruv Yadav
FR - 15	The system shall display a confirmation message after a ticket is successfully submitted.	Dhruv Yadav
FR - 16	The system shall display a dashboard listing tickets submitted by the user	Dhruv Yadav
FR - 17	The system shall allow users to view detailed information for a selected ticket	Dhruv Yadav
FR - 18	The system shall display ticket status indicators such as Open, Assigned, In Progress, and Resolved.	Dhruv Yadav
FR - 19	The system shall allow users to sort tickets by priority within the ticket dashboard.	Dhruv Yadav
FR-20	The ticketing system will archive resolved tickets 2 months after they are resolved and will only be searchable via the 'archived' filter tag.	Nicolas Gaffney
FR-21	The ticketing system will display real time changes when the AI reclassifies ticket severity	Nicolas Gaffney
FR-22	The system will allow admins to configure the AI classification rules and how it determines severity of tickets	Nicolas Gaffney
FR-23	The system shall maintain a complete ticket history	Nicolas Gaffney
FR-24	The system will maintain a log of reclassifications and status changes	Nicolas Gaffney
FR-25	The system will create a weekly report summarizing tickets closed, how fast they were closed, and the team that closed it.	Nicolas Gaffney
FR-26	The system will display a list of tickets that the AI determined is best for the current employee	Nicolas Gaffney
FR-27	There will be a search bar, with the ability to filter based on severity, type of issue, and region	Nicolas Gaffney
FR-28	The system will allow admins to manually reclassify or reassign tickets	Nicolas Gaffney
FR-29	The system will provide a responsive interface accessible from standard web browsers across both desktop and mobile	Nicolas Gaffney

FR-30	The system will relate to the existing data of T-Mobile employee profiles to automatically feed information of users in new tickets.	Rinin Abraham
FR-31	The system will have a REST API feed, to use in real-time data consumption by Tableau dashboards.	Rinin Abraham
FR-32	The system will be flexible to CSV export of weekly data reports.	Rinin Abraham
FR-33	The system will provide automated notifications to the SIEM system within 5 minutes upon identifying deviant access patterns.	Rinin Abraham
FR-34	The system will implement server based session invalidation right after user logout.	Rinin Abraham
FR-35	The system will prevent any SSO reuse upon 8-hour expiry at the API gateway level.	Rinin Abraham
FR-36	The system will enable technicians to make internal notes on the tickets that will not be visible to the end-user.	Rinin Abraham
FR-37	The system will embrace multi-factor authentication (MFA) among the T-Mobile SSO integration.	Rinin Abraham
FR-38	Tickets that have been in the queue of High Severity and had not been assigned to a technician after 30 minutes would be automatically escalated.	Rinin Abraham
FR-39	The system will allow the administrator to view an audit log to monitor all manual ticket overrides.	Rinin Abraham
FR - 40	The System shall periodically monitor for tickets with severity = “High” that are unassigned	Azzam Zahid
FR - 41	The system shall run an escalation check at a configurable interval (every 5 mins) to identify High severity tickets than remain unassigned beyond the configured threshold.	Azzam Zahid
FR - 42	When a ticket is auto-escalate, the system shall set the ticket status to “Escalated” and store an escalated timestamp.	Azzam Zahid
FR - 43	The system shall notify the IT Manager when a ticket is auto escalated.	Azzam Zahid
FR - 44	The system shall create an audit log entry for each auto escalation event containing ticketId, timestamp, old status, and new status.	Azzam Zahid
FR - 45	The system shall provide an “Escalated Tickets” view/filter for IT Support employees.	Azzam Zahid
FR - 46	The system shall allow administrators to configure escalation rules (time threshold, severity threshold, and notification target)	Azzam Zahid
FR – 47	The system shall allow administrators to override escalation status and/or update severity, and record the override in the audit log.	Azzam Zahid
FR - 48	The system shall allow IT support employees to claim or assign an escalated ticket to themselves or another technician	Azzam Zahid

FR - 49	The system shall generate a weekly report summarizing escalations, time to assignment, and tickets most frequently escalated.	Azzam Zahid
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Non-Functional Requirements

Number	Non-Functional Requirements (NFR)	Student/Owner
NFR - 01	The system shall classify incoming tickets within 2 seconds of submission.	Dhruv Yadav
NFR - 02	The system shall support at least 10,000 active tickets without performance degradation.	Dhruv Yadav
NFR - 03	The system shall maintain 99.9% uptime to ensure continuous access for IT personnel.	Dhruv Yadav
NFR - 04	The system shall encrypt data transmitted between system components to protect internal ticket information.	Dhruv Yadav
NFR-05	The interface will match the T-Mobile color scheme on the ticketing page	Nicolas Gaffney
NFR-06	The AI will process and classify tickets in a fast and orderly manner, taking no more than 1 minute to classify a new ticket.	Nicolas Gaffney
NFR-07	The interface will be fully functional on all major web browsers' latest version as of March 5th, 2026 (Chrome, Firefox, Safari, Edge, Opera).	Nicolas Gaffney
NFR-08	The system will abide by T-Mobile privacy policies	Nicolas Gaffney
NFR-09	The system will be available 24/7 with a high uptime rate so that the IT personnel can always access the system.	Rinin Abraham
NFR-10	The AI will be able to process and categorize tickets in a rapid and logical fashion requiring a maximum of 1 minute to categorize a new ticket.	Rinin Abraham
NFR-11	The system will AES-256 encrypt all the data at rest, and access can be logged and audited.	Rinin Abraham
NFR-12	The system will be able to support a minimum of 10,000 active tickets without losing its performance.	Rinin Abraham
NFR -13	The escalation check job shall complete within 10 seconds when evaluating up to 10,000 active tickets	Azzam Zahid
NFR -14	The escalation check job shall run at the configured interval with $\geq 99.9\%$ successful executions per month, and failures shall be logged for review.	Azzam Zahid
NFR-15	The system shall enforce role based access control (RBAC) so that only Admin users can configure escalation rules or override escalation status/severity	Azzam Zahid

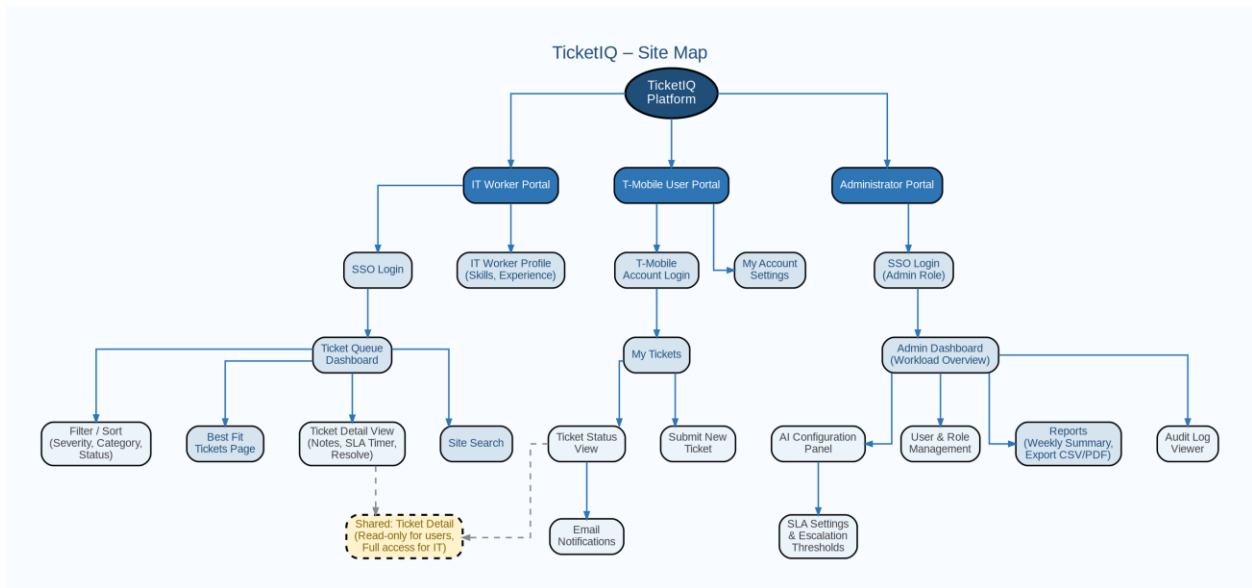
NFR-16	The system shall record each auto escalation and admin override in an audit log and retain audit records for at least 12 months for traceability.	Azzam Zahid
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User stories

Number	User Stories	Student/Owner
US - 01	As a T-Mobile user, I want to submit an IT helpdesk ticket describing my issue, so that the system can analyze the problem and route it to the appropriate IT specialist.	Dhruv Yadav
US - 02	As an IT support employee, I want the system to automatically analyze tickets and determine their urgency and severity, so that critical issues can be identified quickly.	Dhruv Yadav
US - 03	As an IT support employee, I want tickets to be based on urgency and severity, so that I can address the most important issues first.	Dhruv Yadav
US - 04	As an IT support employee, I want the system to assign tickets to technicians based on their expertise and previous work, so that issues are handled by the most qualified person.	Dhruv Yadav
US-05	As an IT manager, I want to generate weekly ticket summaries so that I can view each team's performance and address shortcomings more efficiently.	Nicolas Gaffney
US-06	As an IT worker, I want to filter by ticket severity so that I can resolve the most critical tickets first.	Nicolas Gaffney
US-07	As a T-Mobile User, I want to view my ticket status and classification so that I know the progress on my ticket and don't need to constantly be on call with IT.	Nicolas Gaffney
US-08	As an IT Administrator, I want to be able to manually reclassify and reassign tickets so that I can still maintain control over the system if the AI makes mistakes.	Nicolas Gaffney
US-09	As an Enterprise Architect, I would like the system to run on a multi-zone AWS environment, and to have the pay-off site in case of a regional outage on the infrastructure.	Rinin Abraham
US-10	I would like to have the ability to terminate all user sessions after 8 hours of idle time, as an engineer in charge of security, to ensure that unauthorized access to the system through stale session tokens is blocked.	Rinin Abraham
US-11	I would like to be able to access ticket data as a Data Analyst via an API feed of a REST, or a CSV export, to integrate real time helpdesk metrics into Tableau dashboards to do executive reporting.	Rinin Abraham

US-12	I wish that as a Cloud Operations Engineer, I can have the system raising hard alerts in case AWS spend is at 110% of the monthly base as in this way we can keep the infrastructure cost under tight control.	Rinin Abraham
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Site map

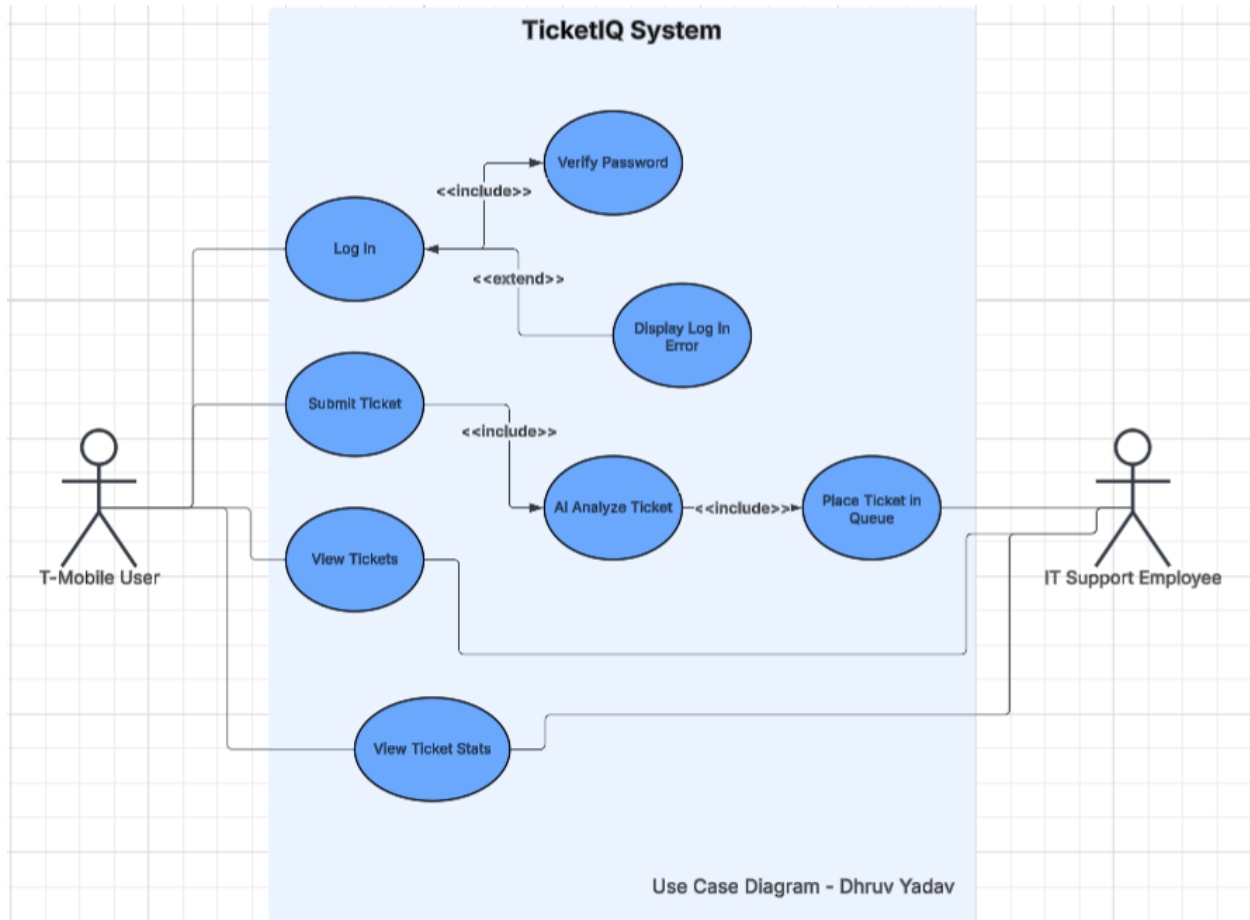


Created by Nicolas Gaffney

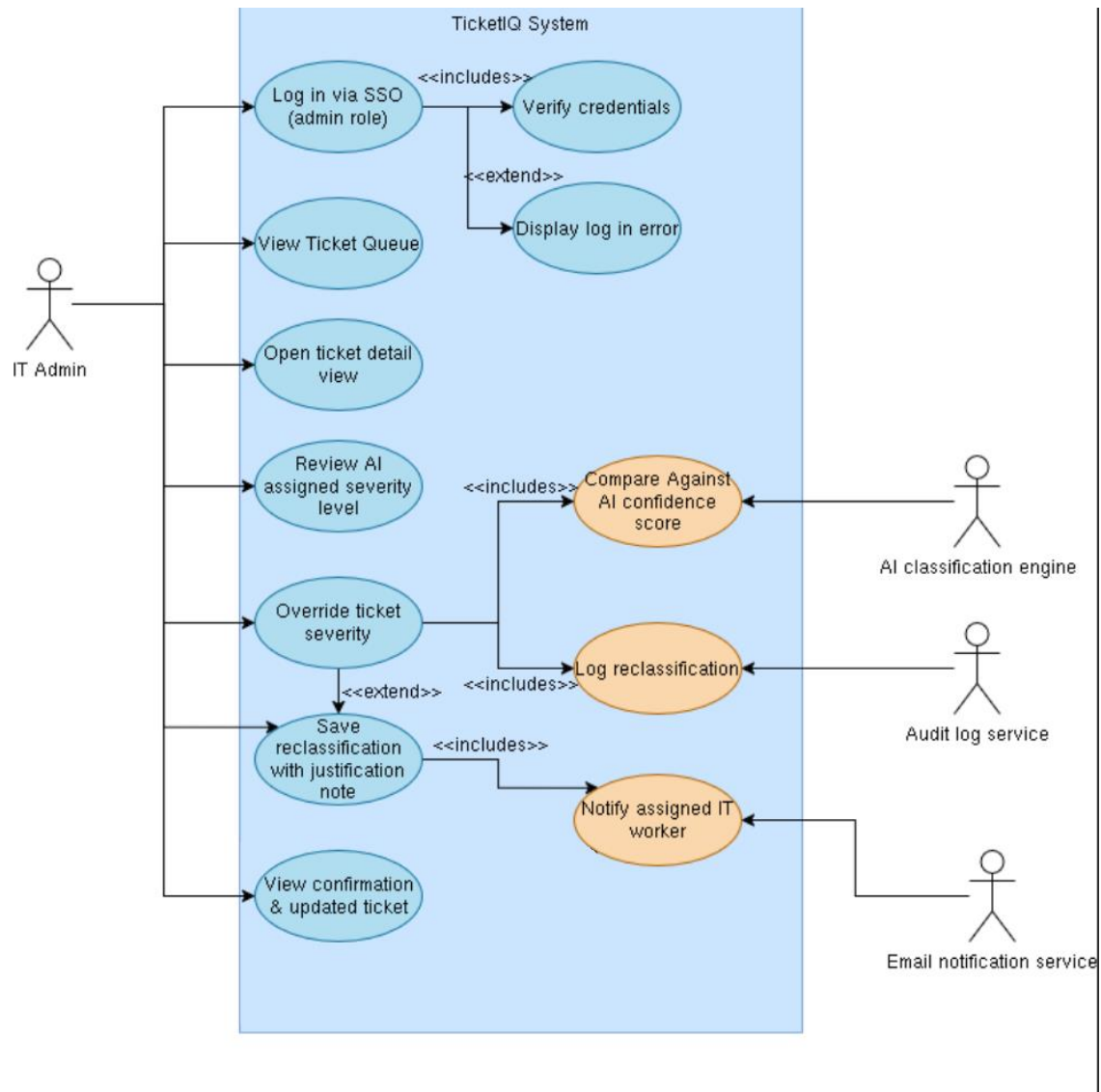
Diagrams

Use Case Diagrams

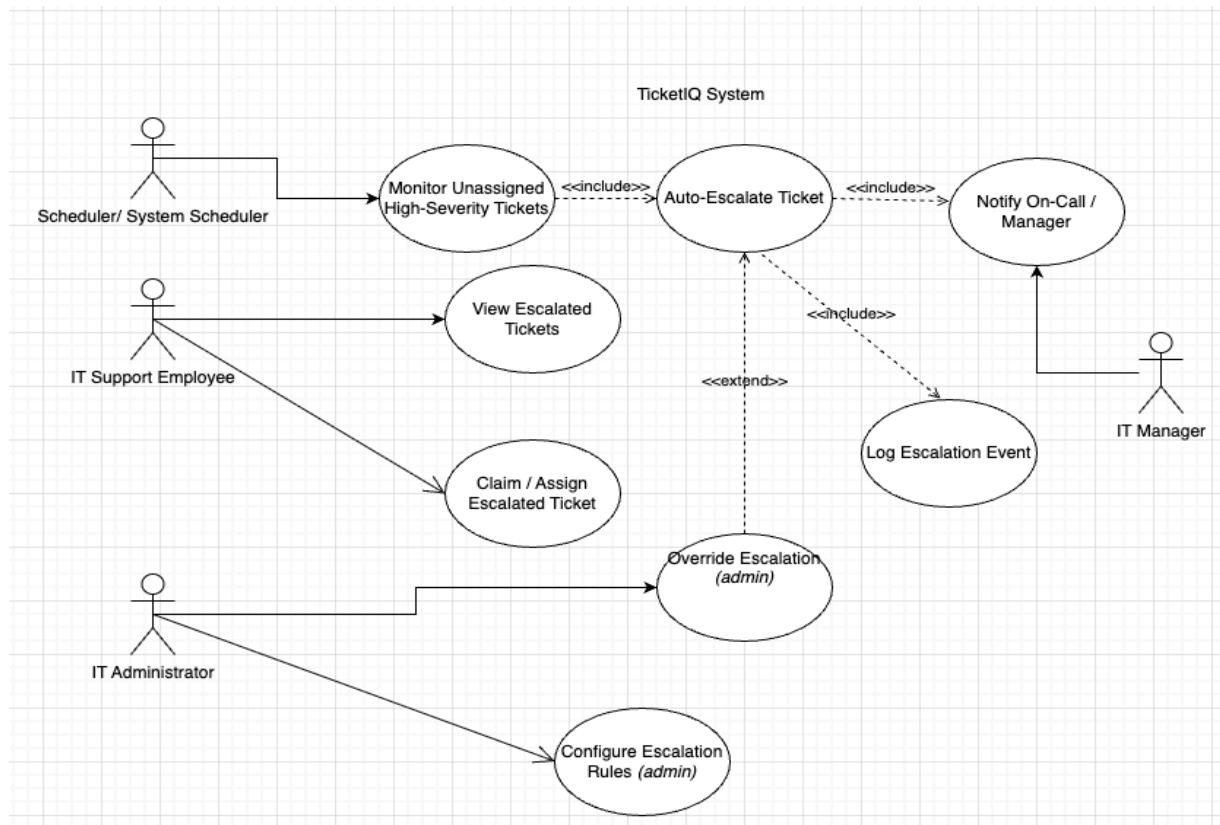
User Interaction with TicketIQ System – Dhruv Yadav



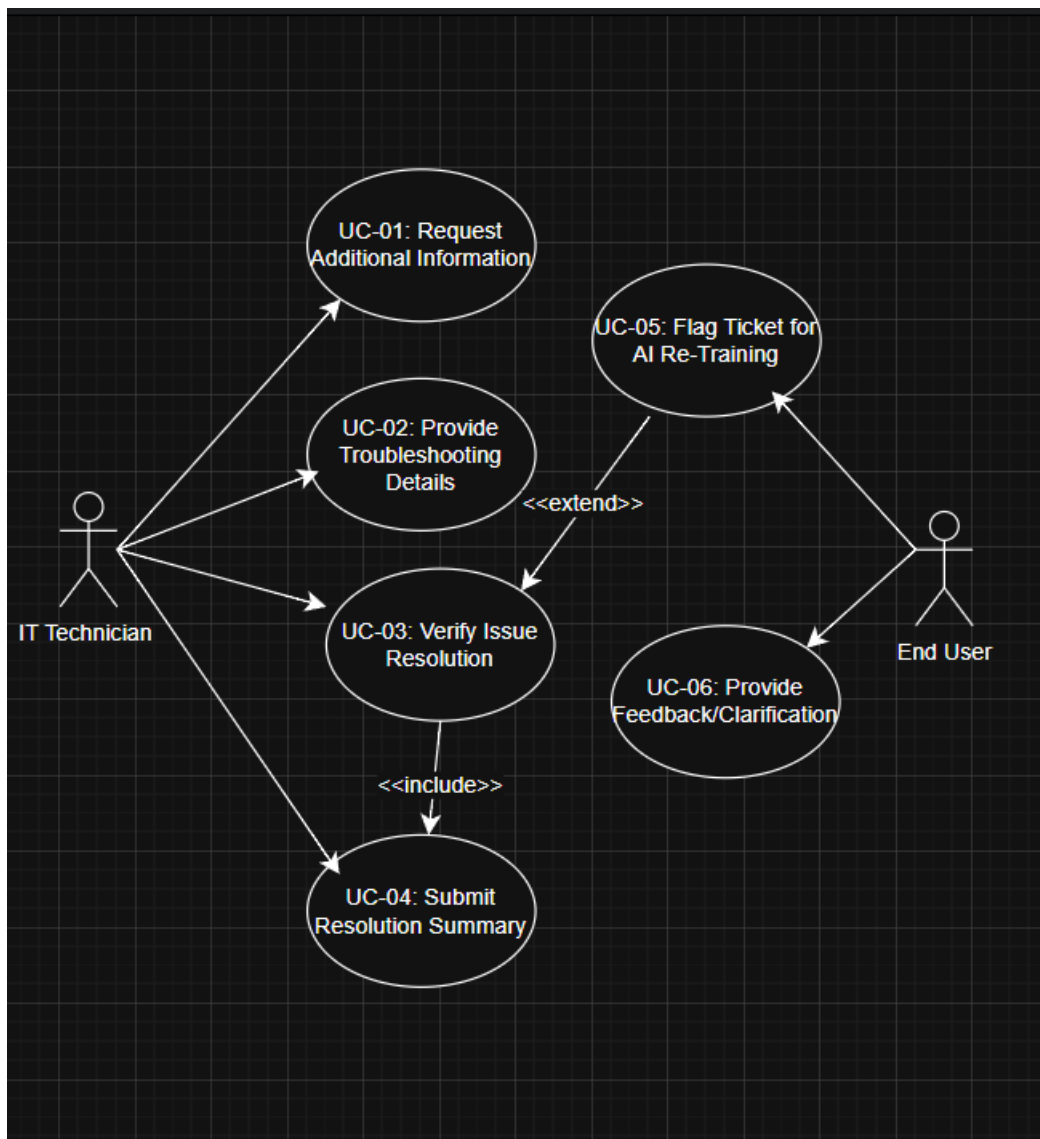
IT admin overrides ticket classification – Nicolas Gaffney



Auto Escalation and Response – Azzam Zahid

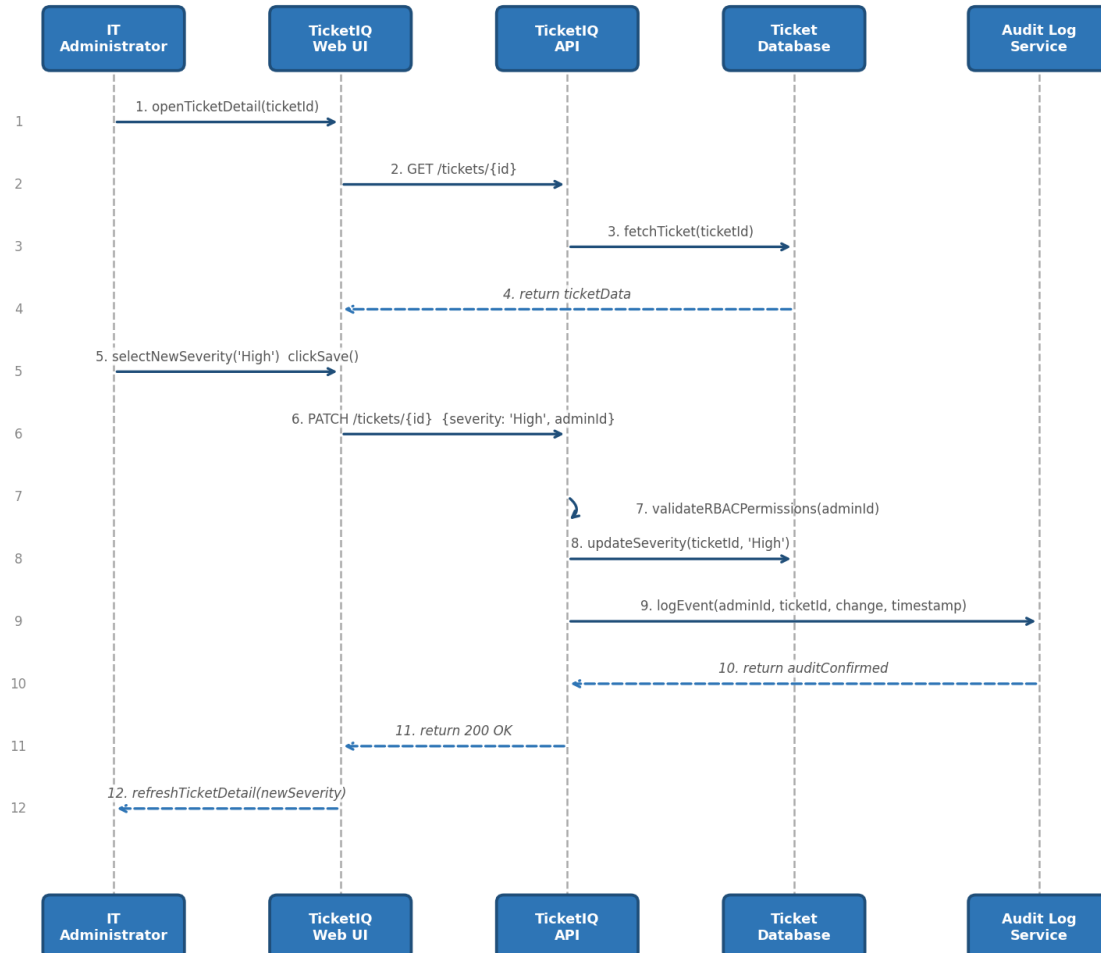


Ticket Resolution & User Feedback – Rinin Abraham



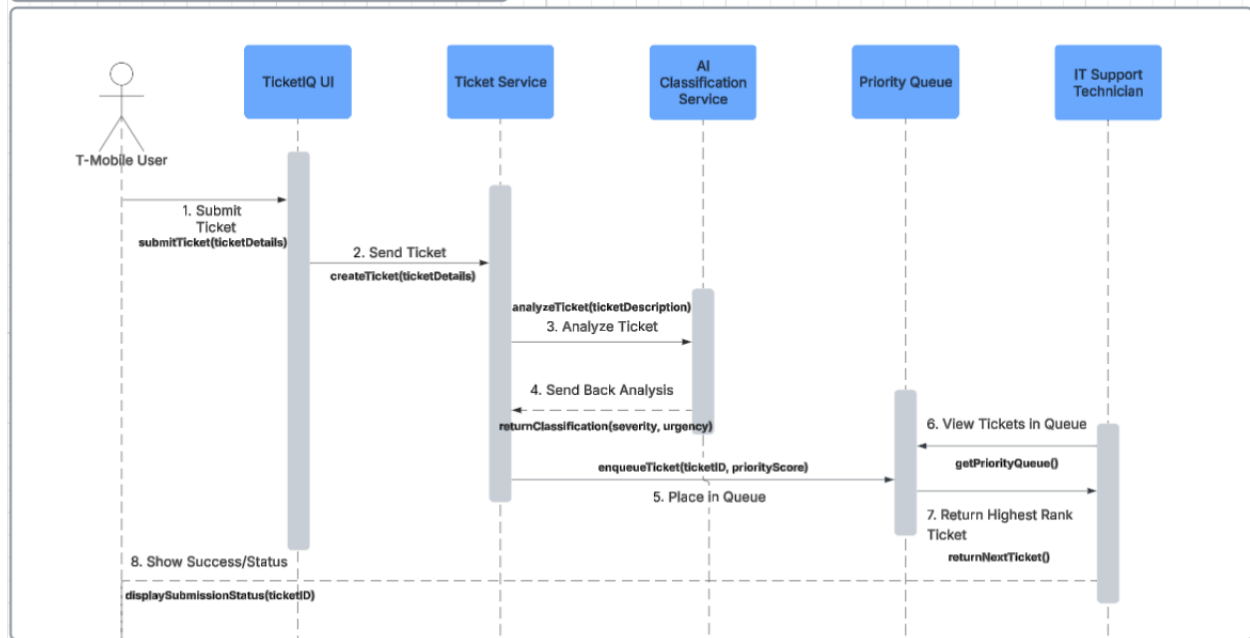
Sequence Diagrams

IT Administrator overrides ticket severity classification – Nicolas Gaffney

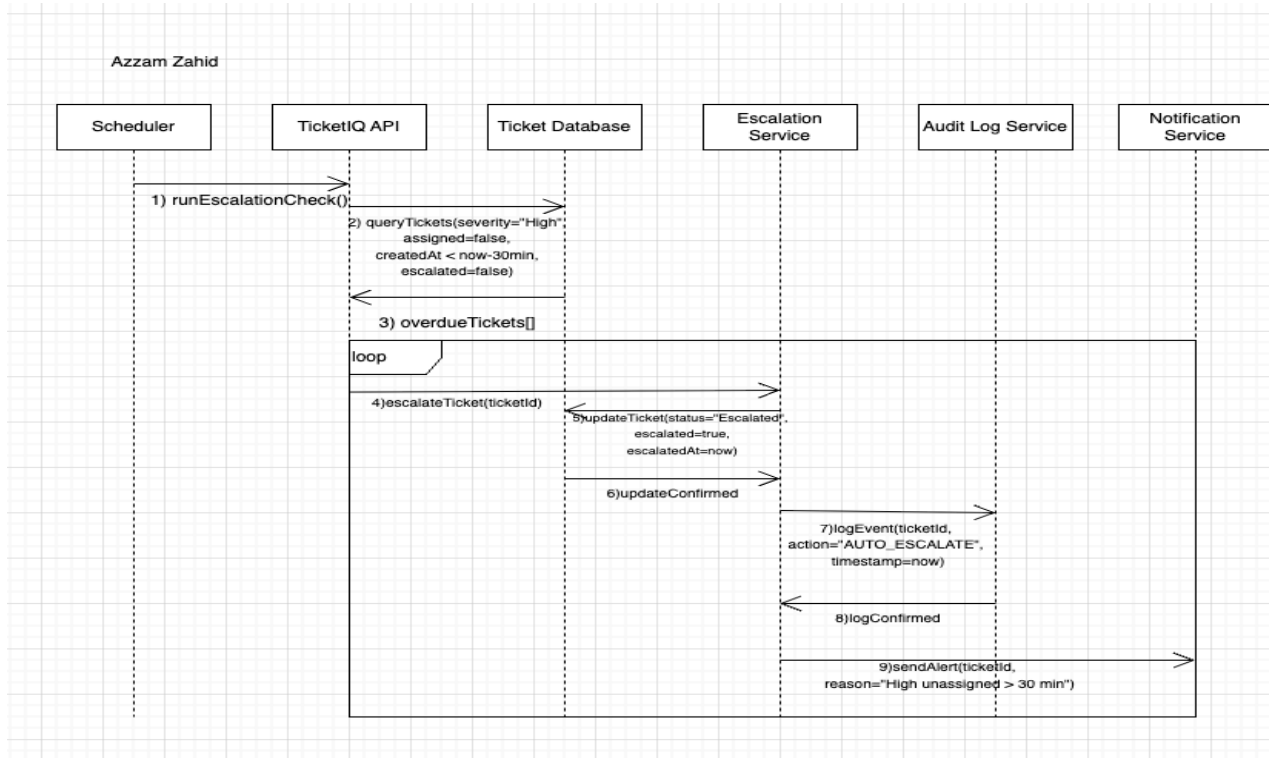


T-Mobile User Submitting a Ticket – Dhruv Yadav

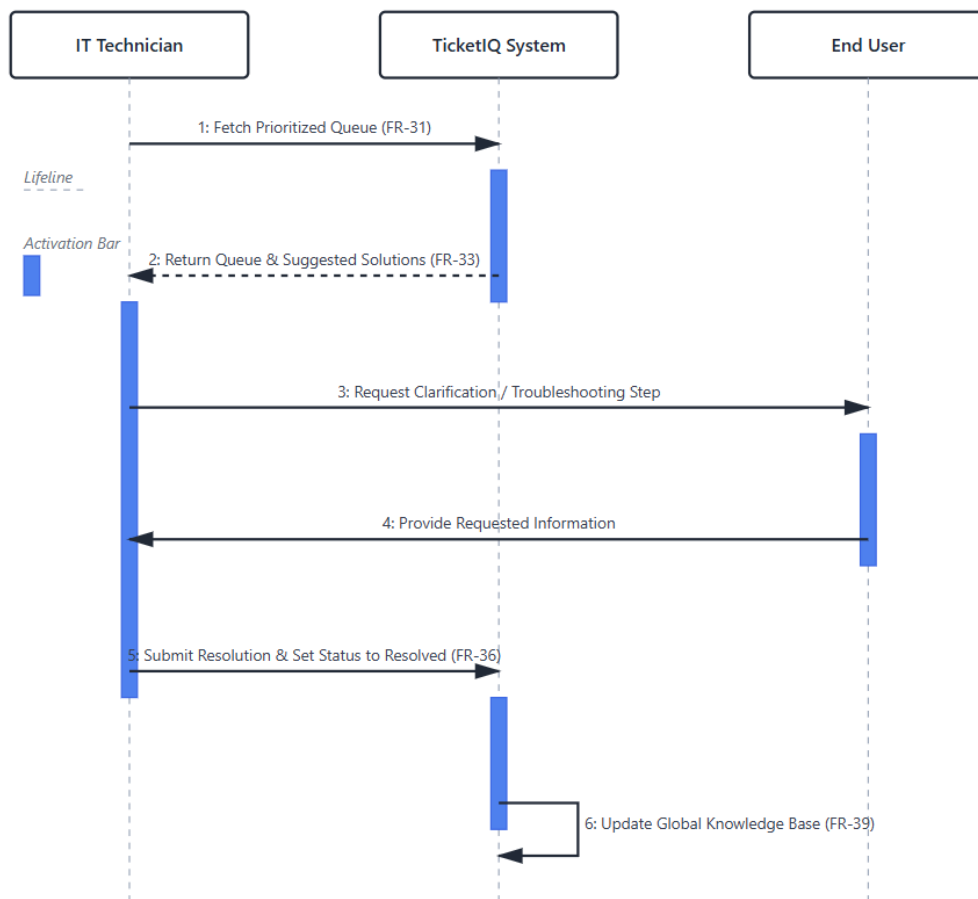
Sequence Diagram - Dhruv Yadav



Auto Escalation after 30 minutes – Azzam Zahid



Technician Resolution and User Feedback – Rinin Abraham



Wire Frames – 03/03/2026

1. Login page – Dhruv Yadav

Wireframe 1 – Login Page (Dhruv Yadav)

TicketIQ Helpdesk System

Login

[Error message area]
e.g., "Invalid username or password"

Username or Email

Password

Login

[Forgot Password?](#)

T-Mobile Internal Support System

2. Submit Ticket Page – Dhruv Yadav

TicketIQ Helpdesk System

[Dashboard](#) [My Tickets](#) [Submit Ticket](#) [Logout](#)

Submit Helpdesk Ticket

Describe the issue you are experiencing so the IT support system can prioritize and route your request.

Ticket Title

Issue Category

[Dropdown] ▼

- Network Issue
- Software Issue
- Hardware Issue
- Account Access
- Other

Priority Level (optional)

[Dropdown] ▼

Issue Description

Attachment Upload

[File Upload Placeholder]

Submit Ticket

Cancel

System Information

Submitted tickets are automatically analyzed by TicketIQ AI to determine urgency and severity. Tickets are then ranked and placed into the IT support priority queue.

T-Mobile Internal IT Support System

3. My Tickets Dashboard (View Tickets) – Dhruv Yadav

TicketIQ Helpdesk System

DashboardMy TicketsSubmit TicketLogout

My Tickets Dashboard

View the status of previously submitted helpdesk tickets and track IT support progress.

[Search Tickets]

Filter: Status

Sort by Priority

Ticket ID	Issue Title	Category	Priority	Status	Assigned IT Technician	Last Updated
1821	VPN Access Issue	Network	High	In Progress	Sarah Kim	March 3
1835	Printer Error	Hardware	Medium	Open	Unassigned	March 2
1844	Email Login Problem	Account	Low	Resolved	Alex Chen	March 1
1815	Software Installation Request	Software	Medium	Assigned	Marcus Lee	March 1

Ticket Details Interaction

Click Ticket → View Ticket Details

- Full description

- Status

- Assigned technician

- Update History

Status Indicators

OpenAssignedIn ProgressResolved

T-Mobile Internal IT Support System

4. Ticket detail view – Nicolas Gaffney

TicketIQ

Ticket Details: TKT-00482

Status: In ProgressSeverity: CRITICALUrgency: HIGH

(FR-18, FR-04, FR-05)

SLA Remaining: 01:24:33 (FR-36)

Description: (FR-02, FR-13)

[ticket description text]

Attachments: screenshot.png (FR-14)

AI Classification: (FR-03, FR-04, FR-05, FR-21)

NLP Severity: CRITICAL | Urgency: HIGH | Priority Score: 98 / 100 | Confidence: 94.2%

Ticket Info: (FR-10)

Submitted by: J. Hernandez | Assigned to: D. Patel | Category: Network | Timestamp: 2026-03-08 09:14

Internal Notes: (FR-36 — not visible to end user)

DP

SL

Add internal note (not visible to user)...

Priority Queue (FR-19)

Queue Position: #1

Severity (FR-28)

CRITICAL

Override Severity

Assignment (FR-28)

D. Patel

Reassign Ticket

Actions

Resolve Ticket

Escalate (FR-38)

Add Internal Note

5. AI Configuration page – Nicolas Gaffney

TicketIQ – AI Configuration (FR-22)

Classification Rules

SLA Settings
Escalation Rules
Routing Weights
Retraining Schedule
Reclassification Log

NLP Classification Rules (FR-03, FR-04, FR-22)

Outage keyword weight 0.85
Security keyword weight 0.90
Minor issue keyword weight 0.25

Severity Score Thresholds (FR-04, FR-05)

High threshold
0.80

Medium threshold
0.50

Low (default)
0.00

Classification timeout (NFR-06)
60 sec

SLA & Escalation Settings (FR-38)

Severity	SLA Limit	Auto-escalate after
High	30 min	30 min unassigned
Medium	4 hrs	—
Low	24 hrs	—

Reset Cancel Save Changes

6. Audit log viewer – Nicolas Gaffney

TicketIQ – Audit Log (FR-39, FR-24)

3,847
Total Events

12
Manual Overrides (FR-28)

5
Escalations (FR-38)

2,901
AI Classifications (FR-03)

Filters: (FR-08, FR-27) Mar 1 – Mar 8, 2026 All Actors All Action Types All Severities

Search keyword... Clear

Timestamp :	Actor	Action (FR-24)	Ticket ID	Details (FR-39)
2026-03-08 09:15	AI Engine	CLASSIFICATION	TKT-00482	NLP severity: HIGH, urgency: HIGH, priority: 98
2026-03-08 09:22	N. Gaffney	OVERRIDE	TKT-00479	Severity: MEDIUM → HIGH. Reason: "Affects 3 depts"
2026-03-08 09:35	N. Gaffney	REASSIGNMENT	TKT-00482	Reassigned: D. Patel → S. Liu
2026-03-08 10:14	D. Patel	CLOSURE	TKT-00477	Resolved: reset AD password via ticket history
2026-03-08 10:28	System	ESCALATION	TKT-00471	Auto-escalated: HIGH unassigned > 30 min (FR-38)
2026-03-08 11:00	M. Chen	LOGIN	—	SSO login · AES-256 session token issued (NFR-11)
2026-03-08 11:15	AI Engine	RECLASSIFICATION	TKT-00485	Real-time severity update: LOW → HIGH (FR-21)

i Audit log retained for 12 months per T-Mobile data retention policy (FR-23, NFR-08). Encrypted at rest per NFR-11.

Showing 1–7 of 3,847 events

7. Escalated Tickets dashboard – Azzam Zahid

Navigation

Dashboard

Escalated Tickets ⭐

All Tickets

Reports

Dashboard

Overview of IT support tickets

Escalated Tickets

8

Require immediate attention

Unassigned

4

Need assignment

Critical Issues

2

Highest priority

Avg Response Time

106m

For escalated tickets

Recent Escalated Tickets

Latest tickets requiring attention

View All

TKT-1007	High	Backup system failed last night	240m ago	US-East	Unassigned	View
TKT-1005	High	SSL certificate expiring in 24 hours	180m ago	Global	Unassigned	View
TKT-1002	High	VPN connection failures for entire department	120m ago	EU-West	Sarah Chen	View
TKT-1004	Critical	Payment gateway API errors	90m ago	US-East	Mike Rodriguez	View
TKT-1008	Medium	Network latency spikes in datacenter	75m ago	EU-West	James Wilson	View

8. Ticket Details Page – Azzam Zahid

Navigation

Dashboard

Escalated Tickets ⭐

All Tickets

Reports

Escalated Tickets ⭐

Critical tickets requiring immediate attention

Search:

Keyword search...

Severity:

All

Status:

Escalated

Region:

All

Age:

All

Sort By:

Age (Oldest)

Showing 8 escalated tickets

Ticket ID	Title	Severity	Status	Age	Assigned To	Actions
TKT-1007	Backup system failed last night	High	Escalated	4h 0m	Unassigned	ClaimAssignOpen
TKT-1005	SSL certificate expiring in 24 hours	High	Escalated	3h 0m	Unassigned	ClaimAssignOpen
TKT-1002	VPN connection failures for entire department	High	Escalated	2h 0m	Sarah Chen	ClaimAssignOpen
TKT-1004	Payment gateway API errors	Critical	Escalated	1h 30m	Mike Rodriguez	ClaimAssignOpen
TKT-1008	Network latency spikes in datacenter	Medium	Escalated	1h 15m	James Wilson	ClaimAssignOpen
TKT-1006	Application server high memory usage	Medium	Escalated	1h 0m	Lisa Park	ClaimAssignOpen
TKT-1001	Database server not responding	Critical	Escalated	45m	Unassigned	ClaimAssignOpen
TKT-1003	Email server delayed messages	High	Escalated	35m	Unassigned	ClaimAssignOpen

9. Admin Escalation and Override – Azzam Zahid

Admin Navigation

Escalation Rules ⭐

Audit Logs

Reports

Escalation Rules Configuration

Configure automatic escalation behavior for support tickets

Escalation Rules

Severity Threshold:

High

Minimum severity to auto-escalate

Time Threshold (minutes):

30

Unassigned time before escalation

Check Interval (minutes):

5

How often system checks for escalation

Notification Target:

IT Manager group

Who receives escalation notifications

Save Configuration

Configuration Version History

Version	Timestamp	Changed By	Changes	Action
v3	2024-03-08 14:23:00	Alex Johnson	Updated time threshold to 30 minutes	View
v2	2024-03-05 09:15:00	Sarah Chen	Changed notification target to IT Manager group	View
v1	2024-03-01 10:00:00	System	Initial configuration	View

Manual Override Panel

Ticket ID:

e.g. TKT-2024-001

Override Severity:

-- No Change --

Override Status:

-- No Change --

Override Reason:

Explain why this override is necessary...

10. Prioritized Technician Dashboard – Rinin Abraham

IT Support

Technician Dashboard

Active Queue

Resolved History

Knowledge Base

Last updated: 1:22:52 AM

Search tickets, users, or issues...

1

Rinin Abraham

Prioritized Ticket Queue

Sorted by AI-generated priority score • 8 active tickets

Ticket ID	Submission Time	User	Issue Category	AI-Priority Score
TKT-2847	09:14 AM	Sarah Chen	Network Connectivity	High Priority 92
TKT-2846	09:08 AM	Michael Roberts	Software Installation	High Priority 88
TKT-2845	08:52 AM	Emily Thompson	Email Access	Medium Priority 67
TKT-2844	08:45 AM	David Kim	Printer Setup	Medium Priority 54
TKT-2843	08:31 AM	Jessica Martinez	Password Reset	Medium Priority 48
TKT-2842	08:22 AM	Ryan Anderson	Hardware Malfunction	High Priority 85
TKT-2841	08:15 AM	Amanda White	VPN Connection	Medium Priority 71
TKT-2840	08:03 AM	James Wilson	Software License	Low Priority 34

Showing 8 tickets

Auto-refresh in 30s

11. Interactive Troubleshooting & User Chat – Rinin Abraham

Awaiting User Input

Communication Hub

TicketIQ Support System

Ticket History

Technical Logs

Ticket #TIQ-2024-0847

In Progress

Email Client Access Issue

Priority: High

Category: Email & Communication

Ticket Created

Ticket

2026-03-09 14:32

User reported unable to access email client

Initial Diagnostic

Technical

2026-03-09 14:35

Network connectivity test passed. Email port 587 accessible.

Assigned to Technician

Ticket

2026-03-09 14:38

Ticket assigned to Rinin Abraham

Authentication Check

Technical

2026-03-09 14:42

Password validation failed. OAuth token expired.

User Contacted

Ticket

2026-03-09 14:45

Live Chat

Request Screen Share

Send Diagnostic Link

End User 14:35

Hi, I can't access my email. It keeps showing an error.

You 14:36

Hello! I'm Rinin, and I'll be helping you today. Can you tell me what error message you're seeing?

End User 14:37

It says "Authentication failed" when I try to log in.

You 14:43

I see. I've run some diagnostics on my end. It looks like your OAuth token might have expired. Let me send you a diagnostic link to check your account

Type your message...

Send

Attach files, images, or diagnostic reports

12. AI Knowledge Base Contribution & Feedback – Rinin Abraham

Knowledge Base Contribution

Ticket #TIQ-2024-0847 - Resolved Successfully

Ticket Summary

Issue Type:

Resolution Time:

Technician:

Email Client Access Issue

18 minutes

Rinin Abraham

AI Classification Review

How accurate was the AI's initial classification of this ticket?

Accurate

AI correctly identified the issue category and priority

Partially Accurate

Some aspects were correct, but needed adjustments

Inaccurate

AI misclassified the issue significantly

Architecture Design – 03/12/2026

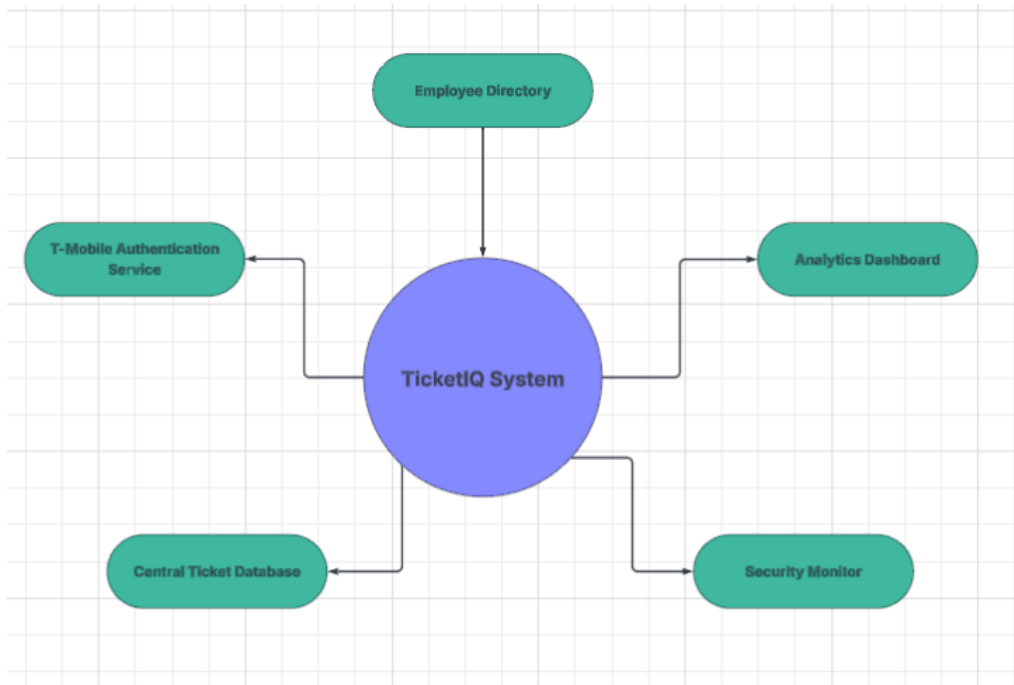
Architecture Strategy

The TicketIQ system uses a layered, distributed architecture that separates functionality into client, security, presentation, business/AI, and data tiers. Users are able to interact with the program through web and mobile interfaces. Moreover, these interactions require authentication before accessing the program's core services. This design supports sustainability and maintainability, aligning with Sommerville's view of using structured architectures to manage and support system evolution.

TicketIQ is structured across three distinct deployment environments –Development, Testing, and Production – each serving a specific role in the software delivery lifecycle. The Development environment is used by team members to implement new features, integrate AI components, and iterate design decisions locally or in isolated cloud workspaces. It is intentionally flexible to support rapid experimentation. The Testing environment mirrors the Production environment as closely as possible, allowing the team to validate system behavior, verify AI classification accuracy, confirm API integrations with T-Mobile's SSO and existing ticketing systems, and conduct security and performance assessments before any release. This environment hosts automated CI/CD pipelines that run regression tests, unit tests, and integration checks on every code commit. The Production environment is the live, end-user-interface deployment of TicketIQ, hosted on scalable cloud infrastructure with strict access controls, monitoring, and rollback capability. Separation of environments ensures that defects are caught before they reach end users and that the system can evolve without disrupting live operations.

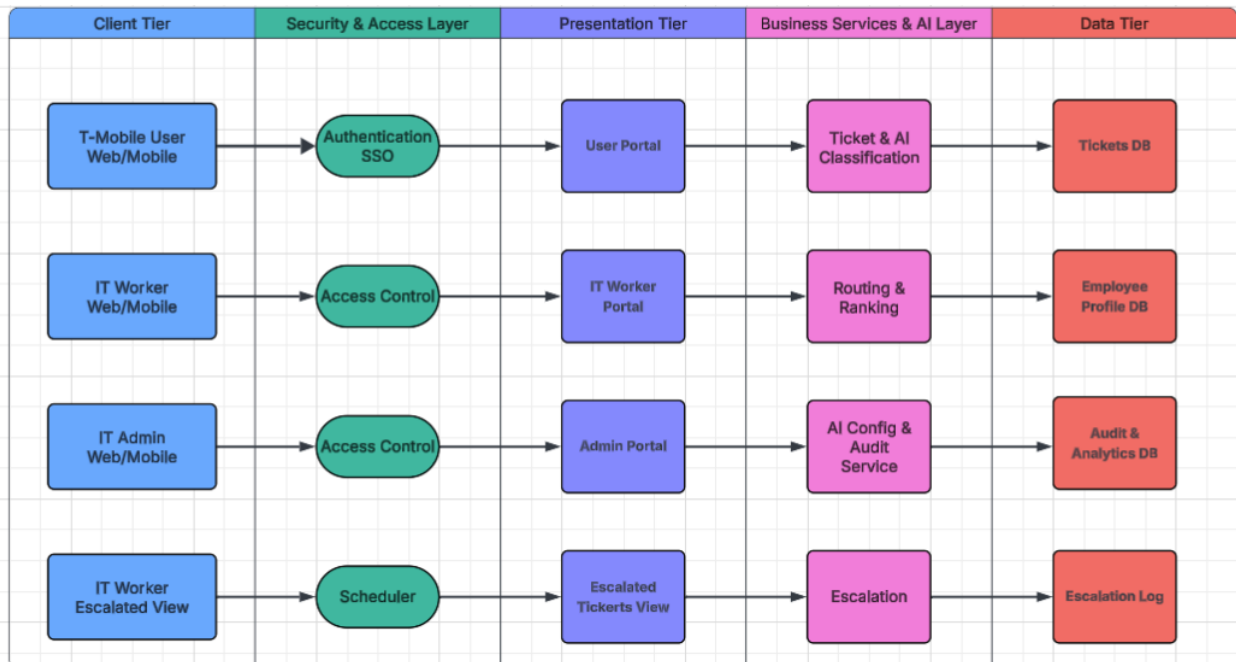
Context Diagram

Dhruv Yadav & Nicolas Gaffney



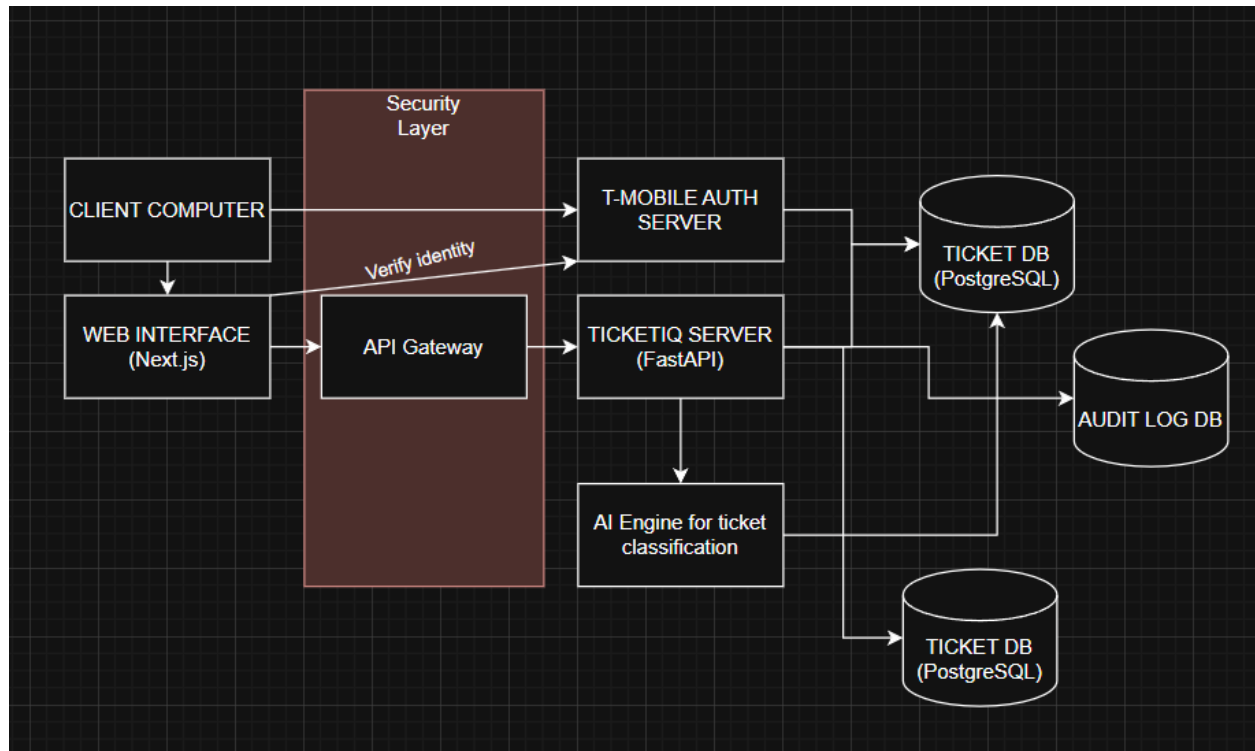
System Architecture (Logical View)

Dhruv Yadav & Azzam Zahid

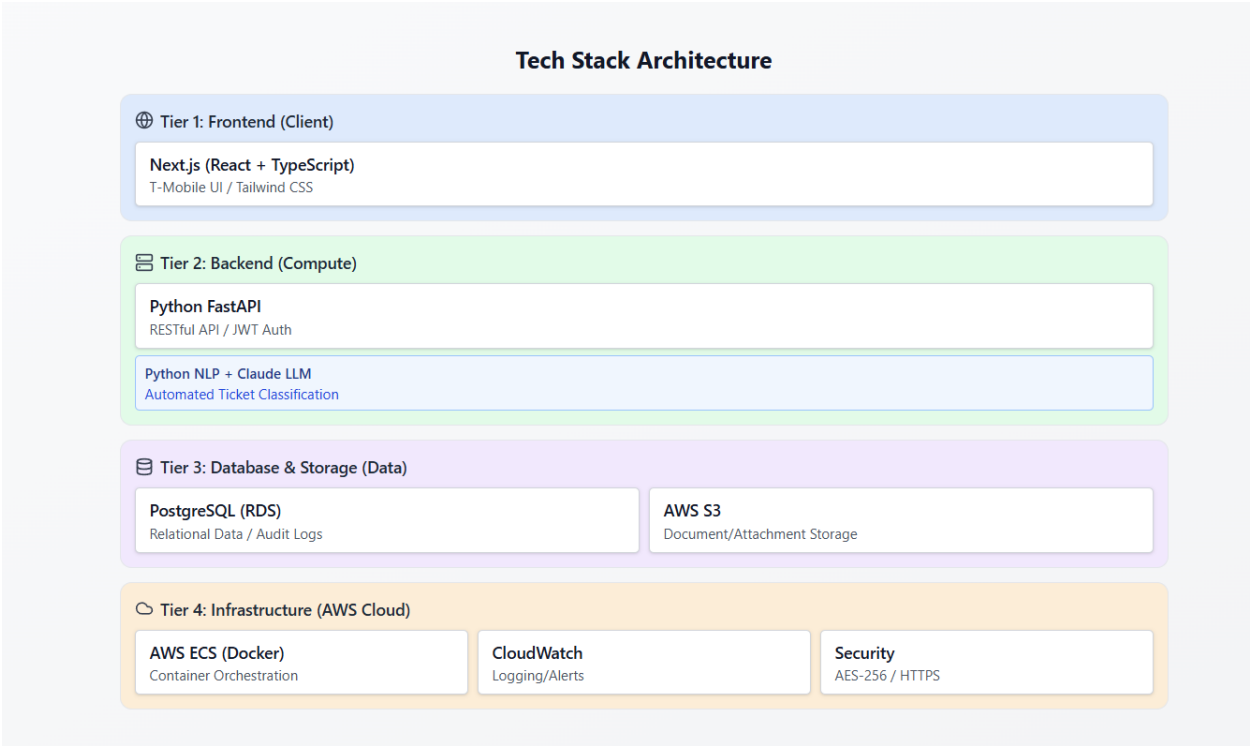


System Architecture (Physical View)

Rinin Abraham & Nicolas Gaffney

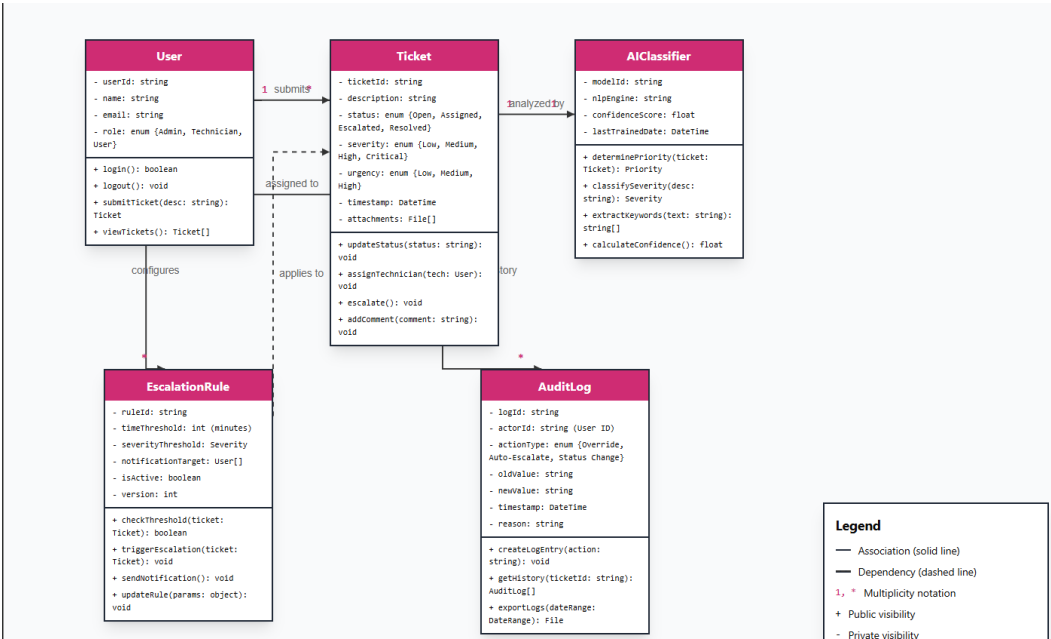


Tech Stack



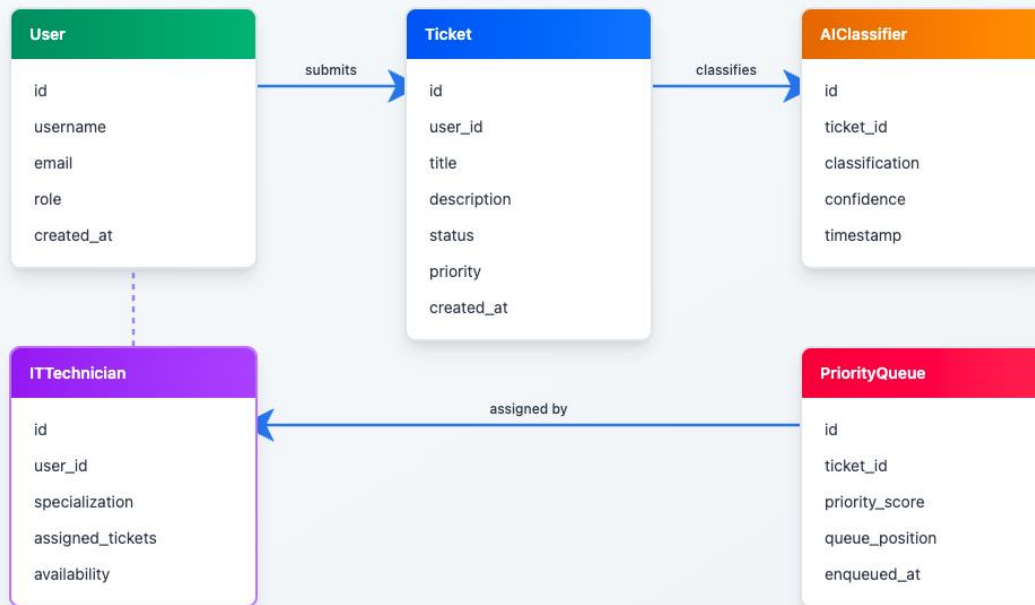
Rinin Abraham

UML Classes and Associations



Rinin Abraham

— Flow/Association - - - Inheritance



Azzam Zahid

Software Testing – 03/05/2026

Testing plan

The testing plan for TicketIQ follows a simple yet structured multilayer approach to ensure most requirements (functional and non-functional) are met by the system prior to delivery. BetterTech will utilize the industry standard unit testing tools for whatever technology is used (Jest for javascript, pytest for python, etc.). The unit tests themselves are to be reviewed by the team lead and 2 senior engineers before it is ever relied upon for testing. Outside of the written tests, a cybersecurity penetration testing team is to be contracted for one week to discover vulnerabilities within the ticketIQ system. AI specific tests are to be created using a custom python testing program to evaluate the specified values and assumptions that the AI makes in regards to whatever test is being performed. All test cases are mapped to their corresponding requirements within the Requirements Traceability Matrix.

The goal is to ensure at minimum passing rate of 95% among the functional requirements and a 90% rate for non-functional requirements. There should be zero critical or high severity security

vulnerabilities, and all medium and low severity are to be documented and scheduled for patching post-launch. All tests regarding the accuracy of AI should have an accuracy of 85% with a target of 93%.

Test cases

Functional test cases

Test Case #	Test Case	Student/Owner
FTC - 01	Login Page Loads Correctly – Go to login page and check that the page displays username/email and password fields, login button, and forgot password link. Result – Login page is correctly displayed	Dhruv Yadav
FTC - 02	User Login is Correct – Enter valid credentials and click Login. Result – User is authenticated and redirected to the dashboard.	Dhruv Yadav
FTC - 03	Invalid Login – Enter incorrect login credentials. Result – Error message is displayed.	Dhruv Yadav
FTC - 04	Submit Valid Ticket – Log in, go to submit ticket page, enter all required information to submit a ticket, click submit. Results – Ticket is successfully created	Dhruv Yadav
FTC - 05	Submit Ticket with Missing Fields - Log in, go to submit ticket page, enter all information to submit a ticket but don't enter information for issue description or category, click submit. Results – Ticket is un-successfully created and errors pop up	Dhruv Yadav
FTC - 06	Select Ticket Category – Navigate to submit ticket page, click issue category dropdown, select a category, enter all other required information, submit. Result – Ticket is submitted and reflected in user end and IT end with correct category	Dhruv Yadav
FTC - 07	Upload Attachment - Navigate to submit ticket page, click upload an image, select an image to upload, enter all other required information, submit. Result – Ticket is submitted and reflected in user end and IT end with submitted image upload.	Dhruv Yadav
FTC - 08	AI Analysis Ticket – Submit a ticket with all correct and required information. Result – Ticket is analyzed and classified with appropriate urgency and severity.	Dhruv Yadav
FTC - 09	Severity Classification (High) - Submit a ticket with all required information and a ticket that should be classified with high severity. Result – AI classifies ticket appropriately with high severity.	Dhruv Yadav

FTC - 10	Severity Classification (Medium) - Submit a ticket with all required information and a ticket that should be classified with medium severity. Result – AI classifies ticket appropriately with medium severity.	Dhruv Yadav
FTC - 11	Severity Classification (Low) - Submit a ticket with all required information and a ticket that should be classified with low severity. Result – AI classifies ticket appropriately with low severity.	Dhruv Yadav
FTC - 12	Urgency Classification (High) - Submit a ticket with all required information and a ticket that should be classified with high urgency. Result – AI classifies ticket appropriately with high urgency.	Dhruv Yadav
FTC - 13	Urgency Classification (Medium) - Submit a ticket with all required information and a ticket that should be classified with medium urgency. Result – AI classifies ticket appropriately with medium urgency.	Dhruv Yadav
FTC - 14	Urgency Classification (Low) - Submit a ticket with all required information and a ticket that should be classified with low urgency. Result – AI classifies ticket appropriately with low urgency.	Dhruv Yadav
FTC - 15	Ticket Prioritization – Submit multiple tickets that will have different classified urgency and severity levels. Result – AI will rank tickets correctly based on urgency and severity and correctly place it in queue.	Dhruv Yadav
FTC - 16	IT Technician Display – Log in as a technician and look at ticket queue. Result – The correct prioritized queue is displayed.	Dhruv Yadav
FTC - 17	Filter Options – Go through each filter option and combination and view results. Result – for every filter option and combination the appropriate tickets are displayed.	Dhruv Yadav
FTC - 18	View Ticket Details – After submitting a ticket, click a ticket from the my tickets page. Result – All details of the ticket shall be displayed correctly.	Dhruv Yadav
FTC - 19	Ticket Status Display – Go to ticket dashboard. Result – Each ticket as an appropriate and correct status displayed (Open, Assigned, In Progress, Resolved) and other metadata saved.	Dhruv Yadav
FTC-20	Ticket archival – Manually add and resolve ticket with an age of 2 months (manually inserting the creation date and resolution date). Refresh the ticket dashboard. Result – Ticket is searchable under the ‘archived’ filter tag on the ticket dashboard.	Nicolas Gaffney

FTC-21	Dynamic Ticket reclassification view – Trigger a reclassification (this can be done by changing weights) on a specified ticket (referred to as ticket T1 from now on) while having the ticket dashboard open and T1 in view. Result – The ticket reclassification happens live on the dashboard	Nicolas Gaffney
FTC-22	Manual ticket reclassification – Log in as an admin. Select a ticket and view details. Override the ticket severity to a different severity than it currently is and hit save. Refresh the ticket dashboard. Result – The ticket is reclassified as the new classification	Nicolas Gaffney
FTC-23	Ticket history completeness – Manually keep track of every ticket submitted in testing in a secondary database. Result – Every ticket made with an age less than 4 years within the history DB is present inside the TicketIQ DB	Nicolas Gaffney
FTC-24	Ticket audit log verification – Log in as an admin, reclassify a ticket, and check the log to ensure reclassification was logged. Log in as IT worker, resolve a ticket, then log in as admin and view log to ensure resolution was logged. Result – The log should accurately represent the changes to the tickets	Nicolas Gaffney
FR-25	Weekly report generation – Create various tickets, and arbitrarily close certain tickets under workers from various teams, manually keeping track of the actions taken under each team. Set the time to Friday at 5PM and allow the system to automatically generate the weekly report. Result – The report should accurately represent the actions recorded by the testing team.	Nicolas Gaffney
FTC-26	Best fit view – Add a dummy employee (E1) to the system with a ticket resolution history filled with a certain category (E1's specialization) of ticket (cellular data issue, home internet, business internet, etc.) and then create two tickets (T1, T2). T1 will be of the same category as E1's specialization, and T2 will not. Result – T1 should be visible under E1's "best fit" page, and T2 should not.	Nicolas Gaffney
FTC-27	Search filter – Create enough tickets to overlap severity, issue type, and region. Use the search bar on the ticket dashboard to filter each of these tags. Result – The search should return 100% of tickets with the specified tag combination and should exclude all tickets not fitting exactly within the parameters set.	Nicolas Gaffney

FTC-28	Ticket reclassification – Log in as an admin. Select a ticket (T1) assigned to an employee (E1) with some severity classification (S1). Update the classification of T1 to some other classification (SX) other than S1. Result – On the ticket dashboard, ticket T1 has classification SX and not S1.	Nicolas Gaffney
FTC-29	Ticket reassignment – Log in as an admin. Select a ticket (T1) assigned to an employee (E1) with some severity classification (S1). Update the assignment of T1 to some other employee (EX) other than E1. Result – While logged in as EX, T1 appears in the best fit page. While logged in as E1, T1 does not appear in the best fit page.	Nicolas Gaffney
FTC-30	Site accessibility – Install the leading web browsers (firefox, chrome) on a desktop and a mobile device. Connect to the local network used for testing and then connect to TicketIQ through each of the web browsers on both devices. Result – The TicketIQ page functions as intended on each web browser on each device.	Nicolas Gaffney
FTC-31	Priority Queue sort - Log into Technician. Make sure that the ticket list has been ranked according to the AI-generated "Urgency Score" (High to Low). Output: Tickets that have the highest score come at the top of the queue.	Rinin Abraham
FTC-32	Opening a particular ticket AI-Suggested Solutions. Click the "AI Insights" panel. Performance: At least 3 relevant past resolutions or Knowledge Base links can be seen on the system through NLP analysis.	Rinin Abraham
FTC-33	Interactive User Chat - Click on the ticket, and go on to the Chat tab and send a message to the End User. Outcome: The message has been sent and shown in End User in his My Tickets view.	Rinin Abraham
FTC-34	Mandatory Resolution Summary - Tries to modify the status of a ticket to Resolved without typing anything in the Resolution Summary box. Outcome: System rejects the change of status and gives a validation error.	Rinin Abraham
FTC-35	Ticket Status Timestamping - Switch a ticket status to be In-Progress. Output: The system logs and shows the precise start time of the technician on the ticket.	Rinin Abraham
FTC-36	Select a Misclassified Ticket Flag - Re-Training AI - Choose a misclassified ticket. Click on the Flag for Accuracy Review button and post. Output: Ticket will be set to an administrative review and not be included in future AI confidence measures.	Rinin Abraham
FTC-37	Knowledge Base Contribution - Solve a ticket and switch on the "Publish to Global Knowledge Base" switch. Outcome:	Rinin Abraham

	The resolution summary has been indexed and can be searched in KB module.	
FTC-38	Saving Draft Resolution - Type into text area and wait 30 seconds or save draft by clicking save draft. Refresh the page. Output: The typed text will be on the editor (NFR-14 Reliability verifying).	Rinin Abraham
FTC-39	Technician Category Filtering - On the Technician Dashboard, dropdown it to select Network Issues. Outcome: The tickets that are visible in the queue are only of the type of Network.	Rinin Abraham
FTC-40	Resolution Attachment Verification - Provide a diagnostic log or a screenshot of a ticket that has been solved and press submit. Outcome: The attachment is stored and can be accessed by both Technician and the Admin as a part of the audit log.	Rinin Abraham
FTC-41	Auto-Escalation Trigger – Create a High severity ticket and leave it unassigned for >30 minutes , then run the escalation check. Outcome: Ticket status changes to Escalated and escalated At timestamp is stored.	Azzam Zahid
FTC42	No Escalation Before Threshold – Create a High severity ticket unassigned and run escalation check at <30 minutes . Outcome: Ticket remains not escalated (status unchanged).	Azzam Zahid
FTC-43	Escalation Notification – Trigger auto-escalation for an eligible ticket. Outcome: On-Call/IT Manager receives a notification containing ticket ID, severity, and reason (“unassigned > threshold”).	Azzam Zahid
FTC-44	Escalation Audit Log Entry – Trigger auto-escalation, then open audit logs for the ticket. Outcome: Audit log records AUTO_ESCALATE with timestamp, old status, new status, and actor=system.	Azzam Zahid
FTC-45	Escalated Tickets Filter View – Ensure at least one escalated ticket exists, then open the queue and apply Status = Escalated filter. Outcome: Only escalated tickets appear in results; non-escalated tickets are excluded.	Azzam Zahid
FTC-46	Claim/Assign Escalated Ticket – As IT Support, open an escalated ticket and click Claim (or Assign to a technician). Outcome: assignedTo updates successfully and assignment is reflected in ticket history.	Azzam Zahid
FTC-47	Admin Override Escalation – As Admin, manually override an escalated ticket’s status/severity and save. Outcome: Ticket updates to overridden values and an ADMIN_OVERRIDE audit log entry is created.	Azzam Zahid
FTC-48	Escalation Job Interval Enforcement – Set escalation check interval to 5 minutes in admin settings, then observe/simulate multiple scheduler runs. Outcome:	Azzam Zahid

	Escalation check executes at the configured interval and each run is recorded in the system/job logs.	
FTC-49	Escalation Rules Configuration Applies – As Admin, change escalation threshold (e.g., 30 → 20 minutes) and save; create a High unassigned ticket aged >20 minutes and run the check. Outcome: Ticket is escalated using the new rule settings and the configuration change is stored (versioned) for traceability.	Azzam Zahid

Non-functional test cases

Test Case #	Test Case	Student/Owner
NFTC - 01	Ticket Classification Performance – Measure time it takes for a ticket to be classified after the ticket is submitted. Result – Ticket should be classified within 2 seconds	Dhruv Yadav
NFTC - 02	System Performance (Heavy Load) - Simulate and submit a large number of tickets (300). Result – System shall handle load and have consistent stable performance.	Dhruv Yadav
NFTC - 03	System Uptime – Monitor system uptime for set period of time. Result – System shall remain up 99.9% of the time.	Dhruv Yadav
NFTC - 04	Data Transit – Check login and ticket submission pipelines. Result – All information submitted is transmitted through encrypted paths.	Dhruv Yadav
NFTC-05	AI Classification speed – Log in as a user. Submit a ticket. Begin a 1-minute timer. Log in as an IT admin and view all tickets, sorting by most recent. Result – The new ticket is viewable, classified, and assigned to an employee before the timer is finished.	Nicolas Gaffney
NFTC-06	Site accessibility extended - (Extension of FTC-30). Same methodology as FTC-30, except testing is now done on more web browsers (firefox, chrome, opera, edge, safari, brave) and on more operating systems and screen sizes (Windows, Mac, Linux, Android, iOS). Result – The system should be functional <i>and</i> responsive in each case.	Nicolas Gaffney
NFTC-07	Color scheme – Cross reference the colors used on TicketIQ's site with the official color palette used by T-Mobile. Result – The colors hex values should be identical.	Nicolas Gaffney
NFTC-08	Technician Dashboard Performance (Usability) - Use a performance profiling tool (such as Chrome DevTools) to measure the time between the time of login to the time of complete rendering of the AI-Prioritized Queue. Consequence: Within a duration of less than 2.0 seconds, the prioritized queue should be interactive entirely (Testing NFR-11).	Rinin Abraham

NFTC-09	Concurrent Technician Load(Scalability): Load-testing tool (such as JMeter) Load-testing tool (such as JMeter) - Scalability - Add 500 IT Technicians and have them use the resolution workspace and submit summaries at the same time. Outcome: The system will have a response time of less than 3 seconds without lockups or crashes of databases (Testing NFR-12).	Rinin Abraham
NFTC-10	Knowledge Base API Modularity (Maintainability) - Make an effort to replace the existing LLM endpoint with a mock API answer without altering the underlying resolution UI software. Output: The mock data is shown in the panel Suggested Resolves which is a success showing modular architecture (Testing NFR-13).	Rinin Abraham
NFTC-11	Auto-Save Resilience (Reliability) - Open a ticket, type a resolution summary and simulate a network disconnection of 60 seconds. Heal relationship and rejuvenate. Output: The text typed before the failure is saved in the local cache or draft database (Testing NFR-14).	Rinin Abraham
NFTC-12	Performance – Escalation Job Runtime – Seed system with 10,000 active tickets , then run the escalation check and time the execution. Outcome: Job completes within the required time limit (e.g., ≤10 seconds) without errors/timeouts.	Azzam Zahid
NFTC-13	Reliability – Scheduled Job Success Rate – Configure escalation job to run every 5 minutes and observe/simulate repeated runs over a long window (e.g., 24 hours). Outcome: ≥99.9% of runs succeed; failures (if any) are logged and visible for investigation.	Azzam Zahid
NFTC-14	Security – RBAC for Admin Actions – Attempt to access escalation rule configuration and override features as a non-admin (IT Support), then repeat as Admin. Outcome: Non-admin access is denied (blocked UI/403); Admin access succeeds; attempts are logged as appropriate.	Azzam Zahid
NFTC-15	Auditability – Log Retention & Retrieval – Generate an auto-escalation and an admin override, then query audit logs by ticket ID and date range. Outcome: Audit records are retrievable and persist according to the retention requirement (e.g., 12 months).	Azzam Zahid

Requirements Traceability Matrix

Functional Requirements Traceability Matrix

Requirement #	Requirement	Test Case #	Test	Student/Owner
FR - 01	The system shall allow T-Mobile users to submit IT helpdesk tickets.	FTC - 04	Submit Valid Ticket – Log in, go to submit ticket page, enter all required information to submit a ticket, click submit. Results – Ticket is successfully created	Dhruv Yadav
FR - 02	The system shall require tickets to include a problem description and affected system or service.	FTC - 05	Submit Ticket with Missing Fields - Log in, go to submit ticket page, enter all information to submit a ticket but don't enter information for issue description or category, click submit. Results – Ticket is unsuccessfully created and errors pop up	Dhruv Yadav
FR - 03	The system shall analyze ticket descriptions using Natural Language Processing (NLP) to determine ticket severity and urgency.	FTC - 08	AI Analysis Ticket – Submit a ticket with all correct and required information. Result – Ticket is analyzed and classified with appropriate urgency and severity.	Dhruv Yadav

FR - 04	The system shall classify tickets into severity levels such as low, medium, and high based on the issue description.	FTC – 09, FTC – 10, FTC – 11	Severity Classification (High, Medium, Low) - Submit a ticket with all required information and a ticket that should be classified with (high, medium, low) severity. Result – AI classifies ticket appropriately with (high, medium, low) severity.	Dhruv Yadav
FR - 05	The system shall automatically determine ticket urgency based on incident type and system impact.	FTC – 12, FTC – 13, FTC -14	Urgency Classification (High, Medium, Low) - Submit a ticket with all required information and a ticket that should be classified with (high, medium, low) urgency. Result – AI classifies ticket appropriately with (high, medium, low) urgency.	Dhruv Yadav
FR - 06	The system shall automatically prioritize tickets based on their severity and urgency classification.	FTC - 15	Ticket Prioritization – Submit multiple tickets that will have different classified urgency and severity levels. Result – AI will rank tickets correctly based on urgency and severity and correctly place it in queue.	Dhruv Yadav
FR - 07	The system shall display a prioritized ticket queue for technicians to view and manage assigned tickets.	FTC - 16	IT Technician Display – Log in as a technician and look at ticket queue. Result – The correct prioritized queue is displayed.	Dhruv Yadav
FR - 08	The system shall allow technicians to	FTC – 17	Filter Options – Go through each filter	Dhruv Yadav

	search and filter tickets using keywords, severity level, or ticket category.		option and combination and view results. Result – for every filter option and combination the appropriate tickets are displayed.	
FR - 09	The system shall authenticate users through T-Mobile's Single Sign-On (SSO) system.	FTC - 02	User Login is Correct – Enter valid credentials and click Login. Result – User is authenticated and redirected to the dashboard.	Dhruv Yadav
FR - 10	The system shall store ticket metadata including severity, urgency, timestamps, and technician assignment for analytics.	FTC - 19	Ticket Status Display – Go to ticket dashboard. Result – Each ticket as an appropriate and correct status displayed (Open, Assigned, In Progress, Resolved) and other metadata saved.	Dhruv Yadav
FR - 11	The system shall provide a login interface allowing T-Mobile users to enter credentials to access TicketIQ.	FTC - 01	Login Page Loads Correctly – Go to login page and check that the page displays username/email and password fields, login button, and forgot password link. Result – Login page is correctly displayed	Dhruv Yadav
FR - 12	The system shall display an error message when login credentials are invalid.	FTC - 03	Invalid Login – Enter incorrect login credentials. Result – Error message is displayed.	Dhruv Yadav

FR - 13	The system shall allow users to select an issue category when submitting a ticket.	FTC - 06	Select Ticket Category – Navigate to submit ticket page, click issue category dropdown, select a category, enter all other required information, submit. Result – Ticket is submitted and reflected in user end and IT end with correct category	Dhruv Yadav
FR - 14	The system shall allow users to upload attachments when submitting a ticket.	FTC - 07	Upload Attachment - Navigate to submit ticket page, click upload an image, select an image to upload, enter all other required information, submit. Result – Ticket is submitted and reflected in user end and IT end with submitted image upload.	Dhruv Yadav
FR - 15	The system shall display a confirmation message after a ticket is successfully submitted.	FTC - 04	Submit Valid Ticket – Log in, go to submit ticket page, enter all required information to submit a ticket, click submit. Results – Ticket is successfully created	Dhruv Yadav
FR - 16	The system shall display a dashboard listing tickets submitted by the user	FTC - 18	View Ticket Details – After submitting a ticket, click a ticket from the my tickets page. Result – All details of the ticket shall be displayed correctly.	Dhruv Yadav

FR - 17	The system shall allow users to view detailed information for a selected ticket	FTC - 18	View Ticket Details – After submitting a ticket, click a ticket from the my tickets page. Result – All details of the ticket shall be displayed correctly.	Dhruv Yadav
FR - 18	The system shall display ticket status indicators such as Open, Assigned, In Progress, and Resolved.	FTC - 19	Ticket Status Display – Go to ticket dashboard. Result – Each ticket as an appropriate and correct status displayed (Open, Assigned, In Progress, Resolved) and other metadata saved.	Dhruv Yadav
FR - 19	The system shall allow users to sort tickets by priority within the ticket dashboard.	FTC - 17	Filter Options – Go through each filter option and combination and view results. Result – for every filter option and combination the appropriate tickets are displayed.	Dhruv Yadav
FR-20	The ticketing system will archive resolved tickets 2 months after they are resolved and will only be searchable via the ‘archived’ filter tag.	FTC-20	Ticket archival – Manually add and resolve ticket with an age of 2 months (manually inserting the creation date and resolution date). Refresh the ticket dashboard. Result – Ticket is searchable under the ‘archived’ filter tag on the ticket dashboard.	Nicolas Gaffney

FR-21	The ticketing system will display real time changes when the AI reclassifies ticket severity	FTC-21	Dynamic Ticket reclassification view – Trigger a reclassification (this can be done by changing weights) on a specified ticket (referred to as ticket T1 from now on) while having the ticket dashboard open and T1 in view. Result – The ticket reclassification happens live on the dashboard	Nicolas Gaffney
FR-22	The system will allow admins to configure the AI classification rules and how it determines severity of tickets	FTC-22	Manual ticket reclassification – Log in as an admin. Select a ticket and view details. Override the ticket severity to a different severity than it currently is and hit save. Refresh the ticket dashboard. Result – The ticket is reclassified as the new classification	Nicolas Gaffney
FR-23	The system shall maintain a complete ticket history	FTC-23	Ticket history completeness – Manually keep track of every ticket submitted in testing in a secondary database. Result – Every ticket made with an age less than 4 years within the history DB is present inside the TicketIQ DB	Nicolas Gaffney

FR-24	The system will maintain a log of reclassifications and status changes	FTC-24	Ticket audit log verification – Log in as an admin, reclassify a ticket, and check the log to ensure reclassification was logged. Log in as IT worker, resolve a ticket, then log in as admin and view log to ensure resolution was logged. Result – The log should accurately represent the changes to the tickets	Nicolas Gaffney
FR-25	The system will create a weekly report summarizing tickets closed, how fast they were closed, and the team that closed it.	FR-25	Weekly report generation – Create various tickets, and arbitrarily close certain tickets under workers from various teams, manually keeping track of the actions taken under each team. Set the time to Friday at 5PM and allow the system to automatically generate the weekly report. Result – The report should accurately represent the actions recorded by the testing team.	Nicolas Gaffney

FR-26	The system will display a list of tickets that the AI determined is best for the current employee	FTC-26	Best fit view – Add a dummy employee (E1) to the system with a ticket resolution history filled with a certain category (E1's specialization) of ticket (cellular data issue, home internet, business internet, etc.) and then create two tickets (T1, T2). T1 will be of the same category as E1's specialization, and T2 will not. Result – T1 should be visible under E1's "best fit" page, and T2 should not.	Nicolas Gaffney
FR-27	There will be a search bar, with the ability to filter based on severity, type of issue, and region	FTC-27	Search filter – Create enough tickets to overlap severity, issue type, and region. Use the search bar on the ticket dashboard to filter each of these tags. Result – The search should return 100% of tickets with the specified tag combination and should exclude all tickets not fitting exactly within the parameters set.	Nicolas Gaffney

FR-28	The system will allow admins to manually reclassify or reassign tickets	FTC-28	Ticket reclassification – Log in as an admin. Select a ticket (T1) assigned to an employee (E1) with some severity classification (S1). Update the classification of T1 to some other classification (SX) other than S1. Result – On the ticket dashboard, ticket T1 has classification SX and not S1.	Nicolas Gaffney
FR-29	The system will provide a responsive interface accessible from standard web browsers across both desktop and mobile	FTC-29	Ticket reassignment – Log in as an admin. Select a ticket (T1) assigned to an employee (E1) with some severity classification (S1). Update the assignment of T1 to some other employee (EX) other than E1. Result – While logged in as EX, T1 appears in the best fit page. While logged in as E1, T1 does not appear in the best fit page.	Nicolas Gaffney

FR-29	The system will provide a responsive interface accessible from standard web browsers across both desktop and mobile	FTC-30	Site accessibility – Install the leading web browsers (firefox, chrome) on a desktop and a mobile device. Connect to the local network used for testing and then connect to TicketIQ through each of the web browsers on both devices. Result – The TicketIQ page functions as intended on each web browser on each device.	Nicolas Gaffney
FR-40	The System shall periodically monitor for tickets with severity = “High” that are unassigned	FTC-41	Auto-Escalation Trigger – Create a High severity ticket and leave it unassigned for >30 minutes, then run escalation check. Outcome: status changes to Escalated and timestamp stored.	Azzam Zahid
FR-41	The system shall run an escalation check at a configurable interval (every 5 mins) to identify High severity tickets that remain unassigned beyond the configured threshold	FTC-48	Escalation Job Interval Enforcement – Set check interval to 5 minutes; observe/simulate scheduler runs. Outcome: check executes on schedule and each run is recorded.	Azzam Zahid
FR-42	When a ticket is auto-escalate, the system shall set the ticket status to “Escalated” and store an escalated timestamp	FTC-41	Auto-Escalation Trigger – Outcome: ticket status becomes Escalated and escalatedAt stored.	Azzam Zahid
FR-43	The system shall notify the IT Manager when a	FTC-43	Escalation Notification – Trigger auto-escalation. Outcome: On-Call/IT Manager	Azzam Zahid

	ticket is auto escalated		receives notification with ticket ID + reason.	
FR-44	The system shall create an audit log entry for each auto escalation event containing ticketId, timestamp, old status, and new status	FTC-44	Escalation Audit Log Entry – Trigger escalation then check logs. Outcome: AUTO_ESCALATE event logged with required fields.	Azzam Zahid
FR-45	The system shall provide an “Escalated Tickets” view/filter for IT Support employees	FTC-45	Escalated Tickets Filter View – Apply Status=Escalated filter. Outcome: only escalated tickets appear.	Azzam Zahid
FR-46	The system shall allow administrators to configure escalation rules (time threshold, severity threshold, and notification target)	FTC-49	Escalation Rules Configuration Applies – Update threshold (30→20), save, run check. Outcome: new rules apply and change is stored/versioned.	Azzam Zahid
FR-47	The system shall allow administrators to override escalation status and/or update severity, and record the override in the audit log	FTC-47	Admin Override Escalation – Override status/severity. Outcome: ticket updates + ADMIN_OVERRIDE audit log entry created.	Azzam Zahid
FR-48	The system shall allow IT support employees to claim or assign an escalated ticket to themselves or another technician	FTC-46	Claim/Assign Escalated Ticket – Claim/assign escalated ticket. Outcome: assignedTo updates and history reflects assignment.	Azzam Zahid
FR-49	The system shall generate a weekly report summarizing escalations, time to assignment, and tickets most frequently escalated	FTC-50	Weekly Escalation Metrics Report – Generate weekly report. Outcome: report includes escalation count +	Azzam Zahid

			time-to-assignment metrics.	
FR-50	The system will be strictly adhering to the T-Mobile brand guide in the frontend, namely magenta and white.	FTC-30	Branding UI Verification: Log in to the portal and view CSS/UI components. Output: UI is equivalent to T-Mobile signature color codes.	Rinin Abraham
FR-51	The helpdesk portal will be compatible with the new Chrome, Firefox, Safari, Edge, and Opera.	FTC-30	Multi-Browser Testing: load the system into all the five browsers. Selection: Portal works and there are no layout breaks.	Rinin Abraham
FR-52	The system will provide updates of classification real-time using WebSockets or JSON over HTTPS.	FTC-31	Priority queue and Ai score	Rinin Abraham
FR-53	The system will cover or conceal PII on logs and reports to conform to T-Mobile privacy standards.	FTC-40	PII Redaction Test: Open a log file or a report that has phone numbers or names of a ticket. Result: PII is masked (e.g., ****).	Rinin Abraham
FR-54	The algorithm will be used to rank the tickets as the Best Fit algorithm will consider the categories of the past resolved by a technician.	FTC-39	Algorithms Test: Assign a ticket Network to a tech. Result Network tickets should now be on the top of their list.	Rinin Abraham
FR-55	The system will have an interactive web interface which will be available on both portable and desktop devices.	FTC-30	Responsiveness Check: Access the dashboard on a mobile phone and a display screen.	Rinin Abraham
FR-56	The system will contain a record of	FTC-35	Ticket timestamps and logs	Rinin Abraham

	every reclassification and status conversion within a period of 12 months.			
FR-57	The system will automate the us-west-2 to the failover in 15 minutes of any outage identification.	FTC-38	Saving Draft Resolution	Rinin Abraham
FR-58	The system will show real time ticket status and AI classification to the end-user.	FTC-31	User Transparency View: Log in as a user and have a My Tickets button. Output: User would be shown the status (e.g. assigned) and AI tag.	Rinin Abraham
FR-59	The system will do a nightly encrypted backup of ticket metadata and AI configuration rules.	FTC-40	Audit log and attachment storage verification	Rinin Abraham

Non-Functional Requirements Traceability Matrix

Requirement #	Requirement	Test Case #	Test Case	Student/Owner
NFR - 01	The system shall classify incoming tickets within 2 seconds of submission.	NFTC - 01	Ticket Classification Performance – Measure time it takes for a ticket to be classified after the ticket is submitted. Result – Ticket should be classified within 2 seconds	Dhruv Yadav
NFR - 02	The system shall support at least 10,000 active tickets without performance degradation.	NFTC - 02	System Performance (Heavy Load) - Simulate and submit a large number of tickets (300). Result – System shall handle load and have consistent stable performance.	Dhruv Yadav
NFR - 03	The system shall maintain 99.9% uptime to ensure continuous access for IT personnel.	NFTC - 03	System Uptime – Monitor system uptime for set period of time. Result – System shall remain up 99.9% of the time.	Dhruv Yadav
NFR - 04	The system shall encrypt data transmitted between system components to protect internal ticket information.	NFTC - 04	Data Transit – Check login and ticket submission pipelines. Result – All information submitted is transmitted through encrypted paths.	Dhruv Yadav

NFR-05	The AI will process and classify tickets in a fast and orderly manner, taking no more than 1 minute to classify a new ticket.	NFTC - 05	AI Classification speed – Log in as a user. Submit a ticket. Begin a 1-minute timer. Log in as an IT admin and view all tickets, sorting by most recent. Result – The new ticket is viewable, classified, and assigned to an employee before the timer is finished.	Nicolas Gaffney
NFR-06	The interface will be fully functional on all major web browsers' latest version as of March 5th, 2026 (Chrome, Firefox, Safari, Edge, Opera).	NFTC - 06	Site accessibility extended - (Extension of FTC-30). Same methodology as FTC-30, except testing is now done on more web browsers (firefox, chrome, opera, edge, safari, brave) and on more operating systems and screen sizes (Windows, Mac, Linux, Android, iOS). Result – The system should be functional <i>and</i> responsive in each case.	Nicolas Gaffney
NFR-07	The system will abide by T-Mobile privacy policies	NFTC - 07	Color scheme – Cross reference the colors used on TicketIQ's site with the official color palette used by T-Mobile. Result – The colors hex values should be identical.	Nicolas Gaffney
NFR-08	The technician dashboard shall ensure all UI components are fully interactive when login	NFTC- 08	Use chrome tools to measure time from logic to complete rendering. Result: Interactive within 3.0 seconds.	Rinin Abraham

NFR-09	The system shall handle a concurrent load of 500 IT technicians performing write operations	NFTC-09	Use JMeter to simulate technicians. Result: response time <4s without database lockups.	Rinin Abraham
NFR-10	The system architecture shall maintain a modular design allowing the replacement of external service endpoints.	NFTC-10	Replace existing LLM endpoint with a mock API. Output: Mock data displays in panel	Rinin Abraham
NFR-11	The system shall provide autosave	NFTC-11	Simulate 60s network disconnection while typing. Output: Text is recovered from local cache	Rinin Abraham
NFR-13	The escalation check job shall complete within 10 seconds when evaluating up to 10,000 active tickets	NFTC-13	Performance – Escalation Job Runtime – Seed 10k tickets; run job; Outcome: completes within time limit.	Azzam Zahid
NFR-14	The escalation check job shall run at the configured interval with $\geq 99.9\%$ successful executions per month, and failures shall be logged for review	NFTC-14	Reliability – Scheduled Job Success Rate – Observe/simulate repeated runs; Outcome: $\geq 99.9\%$ success and failures logged.	Azzam Zahid
NFR-15	The system shall enforce role based access control (RBAC) so that only Admin users can configure escalation rules or override escalation status/severity	NFTC-15	Security – RBAC for Admin Actions – Non-admin blocked; admin allowed; Outcome: access enforced and logged.	Azzam Zahid
NFR-16	The system shall record each auto escalation and admin override in an audit log and retain audit	NFTC-16	Auditability – Log Retention & Retrieval – Query audit logs by ticket/date; Outcome:	Azzam Zahid

	records for at least 12 months for traceability		records retrievable and retained.	
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Glossary of Terms – 03/08/2026

Term	Definition
Ticket	A submitted request for IT help that contains a problem description, metadata (category/region/system), and a status.
Ticket Queue	The ordered list of open tickets waiting to be worked, typically sorted by priority.
Severity	How serious the issue is based on impact (e.g., High/Medium/Low).
Urgency	How quickly the issue needs attention based on time sensitivity.
Priority	A ranking used to order tickets, often computed from severity and urgency.
Escalation	Raising a ticket's attention level when conditions are met (e.g., High severity unassigned beyond a time threshold).
Auto-Escalation	Escalation triggered automatically by the system (not manually by a person).
On-Call	The designated support lead responsible for handling urgent issues during a specific shift.
SSO (Single Sign-On)	Authentication method where users sign in through T-Mobile's identity provider rather than separate app credentials.
RBAC (Role-Based Access Control)	Access control method where permissions depend on the user's role (User, IT Support, Admin).
Admin Override	A manual admin action that changes a system decision (e.g., severity/status/escalation).
Audit Log	A tamper-resistant record of important actions (escalations, overrides, access events) including actor and timestamp.
Ticket History	Timeline of changes to a ticket (status updates, assignments, severity changes, comments).
NLP (Natural Language Processing)	Techniques used to analyze ticket text to infer category, severity, or urgency.
AI Classification	Automated analysis that assigns severity/urgency/category to a ticket based on its content and metadata.

Reclassification	Updating a ticket's classification (severity/urgency/category) after new info is added or the model changes.
Scheduler / Job Runner	Background service that runs recurring system tasks (e.g., escalation checks, weekly reports).
Threshold	A rule value that triggers an action (e.g., 30 minutes unassigned → auto-escalate).
RTM (Requirements Traceability Matrix)	A table that maps each requirement to one or more test cases to prove coverage
Functional Requirement (FR)	A requirement describing a capability the system must provide (what it does).
Non-Functional Requirement (NFR)	A requirement describing a quality attribute (performance, security, reliability, usability).
Functional Test Case (FTC)	A test that validates a functional requirement.
Non-Functional Test Case (NFTC)	A test that validates a non-functional requirement.
SLA (Service Level Agreement)	A target response or resolution standard (e.g., high severity tickets handled within a certain time).